

Research papers

Flood susceptibility prediction in the Indo-China Peninsula using 21 years of inundation occurrence data and machine learning

Gangya Huang^a, Alex Hay-Man Ng^{a,*}, Qianqian Zhou^{a,*}, Qinggaozi Zhu^b, Zheyuan Du^c, Linlin Ge^d

^a School of Civil and Transportation Engineering, Guangdong University of Technology, No. 100 Waihuan Xi Road, Guangzhou 510006, China

^b NSW Department of Climate Change, Energy, the Environment and Water, Parramatta, NSW 2150, Australia

^c Geoscience Australia, Symonston, ACT 2609, Australia

^d School of Civil and Environmental Engineering, University of New South Wales, Sydney, Australia

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ABSTRACT

Accurate flood susceptibility prediction is essential for identifying high-risk areas, supporting disaster preparedness, and mitigating socioeconomic losses. However, traditional hydrological models rely on complex physical parameters and large amounts of data, and although they have been used in small-scale flood event simulations, there are still significant research gaps in large-area flood prediction with strong spatial heterogeneity. These challenges are particularly pronounced in data-scarce and hydrologically complex regions such as the Indo-China Peninsula, where historical flood records are sparse and flood events exhibit strong spatial discontinuity. To address these challenges, this study develops a scalable, interpretable machine learning framework for flood susceptibility prediction across transboundary river basins, integrating remote sensing data with environmental variables grounded in water cycle theory. Using 21 years (2003–2023) of daily MODIS imagery at 250 m resolution, we developed a dynamic flood extraction method based on an adaptive Otsu algorithm with temporal aggregation, generating a spatially continuous inundation occurrence dataset. This dataset was used to train an Extreme Gradient Boosting (XGBoost) model incorporating twelve climate–land surface predictors, including Normalized Difference Vegetation Index (NDVI), land use, temperature, elevation, and groundwater anomalies. Model evaluation showed high predictive accuracy ($R^2 = 0.82$, RMSE = 0.02), outperforming other common machine learning and deep learning methods in both performance and efficiency. Explainable artificial intelligence (AI) techniques, SHapley Additive exPlanations (SHAP) and partial dependence plots (PDPs), revealed NDVI, land use and temperature as dominant predictors, highlighting the significance of vegetation–climate interactions in regional flood susceptibility. Partial dependence analysis revealed a nonlinear “heat suppression” threshold. Cropland and urban areas exhibited greater sensitivity to temperature extremes, underscoring the influence of land surface changes on flood dynamics. The 21 years daily average inundation occurrence rate maps produced in this study support a range of applications, including early warning systems, infrastructure planning, and climate adaptation strategies, particularly in data-limited transboundary contexts.

1. Introduction

Flooding is one of the most frequent and devastating natural disasters globally, significantly impacting socio-economic activities and environmental stability (Adhikari et al., 2010; Tellman et al., 2021). Accurate flood susceptibility prediction is essential to minimize these impacts, particularly in flood-prone regions such as the Indo-China Peninsula (ICP), where complex topography and monsoon-dominated

climates frequently lead to large-scale inundation events (Ahamed and Bolten, 2017; Chen et al., 2020; Hu et al., 2018; Liu et al., 2018).

Traditionally, hydrological models such as SWAT, HEC-HMS, VIC, MIKESHE have been widely used for flood forecasting and water resource management (de Paula Netto et al., 2024; Devia et al., 2015; Hosseiny, 2021). Although physically based, these models remain resource-intensive and are often impractical for real-time and large-scale applications due to their high computational demands and

* Corresponding authors.

E-mail addresses: hayman.ng@gdut.edu.cn (A.H.-M. Ng), qiaz@foxmail.com (Q. Zhou), esther.zhu@dcceew.nsw.gov.au (Q. Zhu), zheyuan.du@ga.gov.au (Z. Du), l.ge@unsw.edu.au (L. Ge).

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extensive input data requirements. Moreover, these models are typically calibrated at the catchment scale and rely on detailed physical parameters, including land use, soil properties, and meteorological forcing. However, their transferability and performance are often limited in heterogeneous or transboundary regions with sparse data (Han and Morrison, 2022; Kumar et al., 2023; Noor et al., 2022; Pham et al., 2021). Classical statistical models — including the Gumbel distribution, the Generalized Extreme Value (GEV) distribution, the Log-Pearson Type III distribution (Water Resources Council, 1975), and the peak-over-threshold (POT) method coupled with the Generalized Pareto Distribution (GPD) (Coles et al., 2001) — estimate flood magnitude associated with different return periods using long-term annual maxima or exceedance series. Model calibration is particularly time-consuming and often site-specific, which hampers model transferability across regions or temporal scales. Many of these models assume spatial homogeneity within catchments and rely on discrete gauge observations, making them less effective for capturing the spatial variability of flood dynamics across diverse landscapes (Devia et al., 2015).

Additionally, limited observational records and data discontinuities present major challenges for reliable design flood estimation, particularly in ungauged or data-scarce regions (Boni et al., 2007; Gao et al., 2017). The growing complexity of climate–land surface interactions, compounded by sparse and irregular observations, further limits the models' capability to capture multi-year flood dynamics. Recent advances in flood prediction have shifted toward data-driven approaches that can learn spatial–temporal patterns from multi-source observations (Sanjay Shekar and Vinay, 2021).

In contrast, recent advances in remote sensing and machine learning (ML) offer promising alternatives for large-scale flood modeling under data-limited conditions. Satellite-based sensors, including optical (Donchyts et al., 2016; Ji et al., 2015; Pekel et al., 2016), radar (Amitrano et al., 2018; Yan et al., 2015), and microwave instruments, have been widely used to detect flood extent and generate inundation maps over time. MODIS data, in particular, provide consistent and frequent coverage at 250 m resolution, offering advantages for multi-year, large-area flood susceptibility analysis (Kuenzer et al., 2015).

ML models have been increasingly applied to flood predictions (Fu et al., 2022; Razavi-Termeh et al., 2025; Shen, 2018), often using static indicators such as elevation, slope, and land use to predict flood-prone zones (Abijith et al., 2025; Antzoulatos et al., 2022; Assouline et al., 2024; Avand et al., 2021; Bui et al., 2019; Tien Bui et al., 2020). Among ML techniques, tree-based models — such as random forest (RF) and Extreme Gradient Boosting (XGBoost) — have shown strong predictive performance and scalability for geospatial tasks (Bui et al., 2019; Huang et al., 2019; Kaiser et al., 2022; Lyu and Yin, 2023a; Tien Bui et al., 2020). Despite these advancements, most ML-based flood studies still rely on sparse and event-based flood datasets—either manually curated from historical records or extracted from disaster databases (Kumar et al., 2023). These datasets lack spatial and temporal continuity required for robust flood risk assessment, limiting model generalizability and the ability to assess flood risk under changing climate and land conditions.

In addition, most existing flood detection products rely on single-event imagery or sparse records of historical flood occurrences, which inherently lack the temporal continuity required for analyzing multi-year spatiotemporal inundation patterns, limiting their utility in regional planning and risk assessment, particularly in data-scarce regions (Bou et al., 2024; Misra et al., 2025).

To address these limitations, this study integrates 21 years of satellite observations with explainable ML, develops a scalable flood susceptibility prediction framework at the intersection of multi-year inundation exposure analysis, land–climate interactions, and interpretable hydrological modeling. By applying an adaptive Otsu thresholding algorithm with temporal aggregation, a 21-year time series (2003–2023) of daily MODIS imagery at 250 m resolution is used to generate a consistent and high-quality daily average inundation occurrence rate (DAIOR) dataset

across the ICP that captures the spatial and temporal dynamics of flood occurrence under varying environmental conditions. Here, inundation occurrence refers to the empirical frequency of satellite-observed surface water presence at each pixel and should not be interpreted as hydrological flood-frequency or return-period estimation. Unlike previous work that uses isolated flood events, this approach captures spatially and temporally continuous flood patterns, offering a more robust training foundation. This continuous dataset is then used to train the ML model incorporating twelve climate–land surface predictors, including NDVI, land use, topographic attributes, temperature, and groundwater anomalies (GWA), etc.

The novelty of this study lies in four key contributions. First, we construct a regionally consistent inundation dataset derived from daily MODIS imagery, offering a new data resource for flood risk modeling. Second, we design an interpretable machine learning framework that explicitly couples land surface and climate variables within a hydrologically informed modeling approach. Third, we apply explainable Artificial Intelligence (AI) techniques — such as SHAP values and partial dependence plots—to identify the most influential predictors of flood susceptibility and to provide transparent insight into model behavior. Finally, we compare the XGBoost model against other commonly used ML and deep learning (DL) algorithms, including RF, Support Vector Regression (SVR), and Light Gradient Boosting Machine (LightGBM), Long Short-Term Memory networks (LSTM), and Convolutional Neural Networks (CNN), to assess comparative performance and robustness.

By integrating satellite observation, climate–land surface coupling, and interpretable machine learning, this work aims to advance satellite-based flood susceptibility mapping by focusing on observed inundation occurrence rather than return-period-based flood-frequency metrics, while improving interpretability under data-limited conditions.

2. Materials and study area

2.1. Data source

Three major data groups were used in this research: flood influencing data, satellite imagery, and flood events datasets.

2.1.1. Flood influencing data

This study collected flood influencing datasets including meteorological data (precipitation, temperature, vapor pressure difference and wind speed), terrain data (Digital Elevation Model (DEM), slope, Topographic Wetness Index (TWI)), socioeconomic data (landcover, buildings) and environmental indices such as NDVI (Normalized Difference Vegetation Index), GWA (groundwater anomalies) and MSI (moisture stress index). Slope and aspect were derived from the DEM, while TWI was calculated using slope and Global Hydrography Datasets to reflect surface runoff and water aggregation in a basin (Lyu and Yin, 2023b). MSI is a common moisture stress index that can be used to assess the moisture status and wetness of vegetation and soil, and the MSI data were calculated from the SWIR band and NIR band of MODIS. These data description and sources can be found in Table 1.

2.1.2. Satellite images

MODIS data from NASA's Terra and Aqua satellites were used to extract flood-related features. MODIS has had consistent daily coverage since February 2000 and twice-daily coverage since February 2001. We used the MODIS surface reflectance products MOD09GQ/MYD09GQ (collection version 5, L2G) containing the NIR channel (841–876 nm) and the red channel (620–670 nm) at a spatial resolution of 250 m. Terra based datasets (MOD) are available since February 2000, whereas Aqua based datasets (MYD) are available since July 2002. Although the data went through atmospheric correction, atmospheric influences might in some cases not be fully eliminated but only minimized. MODIS is an optical satellite commonly used for flooding mapping and is freely available.

Table 1

Description of predictor variables used in the flood susceptibility prediction.

Factor	Resolution	Description	Data source
NDVI	500 m	normalized difference vegetation index	MOD13A1 V6.1 product
Landcover	10 m	—	Dynamic World V1
DEM	30 m	Digital Elevation Model	NASA SRTM Digital Elevation 30 m
TWI	30 m	Topographic Wetness Index	NASA SRTM Digital Elevation 30 m
Slope	30 m	—	NASA SRTM Digital Elevation 30 m
GWA	55660 m	groundwater anomalies	GRACE mission
MSI	500 m	moisture stress index	MOD09A1.061 product
Wind speed	10 m	—	MOD13Q1.061 product
VPD	4638.3 m	vapor pressure difference	MOD13Q1.061 product
Temperature	1000 m	—	MOD11A2 V6.1 product
Precipitation	5566 m	—	CHIRPS
Buildings	100 m	—	GHSL: Global building volume

2.1.3. Flood events

Three different flood-related datasets were used in this study, each serving a distinct purpose:

Satellite-derived Inundation Occurrence Dataset is generated in this study from 21 years of MODIS surface reflectance imagery and serves as the primary dataset for multi-year inundation occurrence analysis and machine-learning-based flood susceptibility prediction. It provides pixel-level inundation occurrence rate across the entire ICP for the full period 2003–2023.

Global Flood Database (GFD) provides spatially explicit flood inundation polygons derived from MODIS for 913 events between 1989 and 2018 (Tellman et al., 2021). Although its temporal extent ends in 2018, it is used exclusively for event-level validation, enabling direct comparison between our MODIS-based inundation detection and an established global flood product produced using a similar satellite sensor.

The Dartmouth Flood Observatory (DFO) global active archive of large flood Events (1985–present) archive contains authoritative records of major global flood events. In this study, we retrieved 63 flood events in the ICP from quality-controlled DFO records available up to 2018. These events were used to identify major historical floods, guide the selection of MODIS scenes (a total of 12,719 images) for validation, and cross-compare inundation footprints with GFD event polygons.

Notably, although the GFD and DFO datasets have shorter temporal coverage, they are used solely for validation and event identification and do not constrain the temporal extent of the primary satellite-derived inundation dataset.

2.2. Study area

The ICP, encompassing Thailand, Vietnam, Cambodia, Laos, Myanmar, and parts of Peninsular Malaysia, is characterized by diverse topography and a dense network of major rivers, including the Mekong and Chao Phraya. The region lies within the tropical monsoon zone and experiences pronounced seasonal rainfall between May and October, significantly elevating the risk of flooding.

Floods are predominantly concentrated in low-lying basins such as the Mekong Delta and Tonle Sap, whereas high-altitude regions experience considerably fewer events. This spatial pattern, as illustrated in (Fig. 1), highlights the strong topographic control on flood occurrence across the Indo-China Peninsula. Such uneven distribution poses significant challenges to agriculture-based economies, ecosystem stability, and regional water security. Given the transboundary nature of the river systems and the growing vulnerability to climate-induced extremes, accurate and timely flood prediction is vital for disaster risk reduction, sustainable development, and regional water governance.

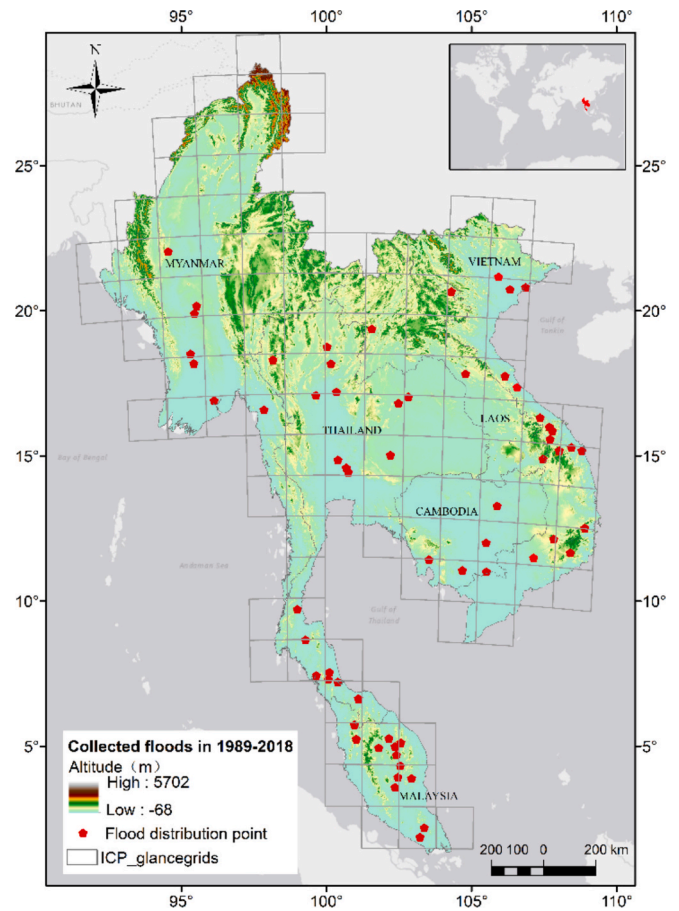


Fig. 1. Distribution and topographic map of flood events in the ICP from 1989 to 2018 obtained from the DFO flood database. The red dots represent the approximate location of historical flood events, and the gray squares represent the mesh blocks used in the inundation extraction algorithm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3. Methodology

Google Earth Engine (GEE) is a cloud-based geospatial analysis platform that provides access to a wide range of satellite imagery and Earth observation datasets, along with powerful processing capabilities. It is widely adopted by researchers and practitioners for mapping and analyzing changes in the Earth's surface. GEE can automatically handle data acquisition, radiometric calibration, and geometric corrections (Kazemi Garajeh et al., 2023; Kordi and Yousefi, 2022). In this study, various MODIS and Landsat global surface products were utilized via the GEE platform. Using this platform, we mapped and monitored flood extent across the ICP from 2003 to 2023. Flood impact data for the same period were also integrated within GEE to support analysis and modeling.

The overall methodology of this study involves multiple stages, including data preprocessing, feature extraction, and machine learning-based flood susceptibility prediction using multi-source datasets. An overview of the methodological framework is illustrated in Fig. 2.

3.1. Inundation extraction algorithm

To extract flood inundation areas, we utilized MODIS surface reflectance products (MOD09GA and MOD09GQ), available on the GEE platform (Gorelick et al., 2017). These products have undergone atmospheric correction (Vermote et al., 2015). The MODIS 09GQ provides surface reflection data in the red band (band 1; 621–670 nm) and near-

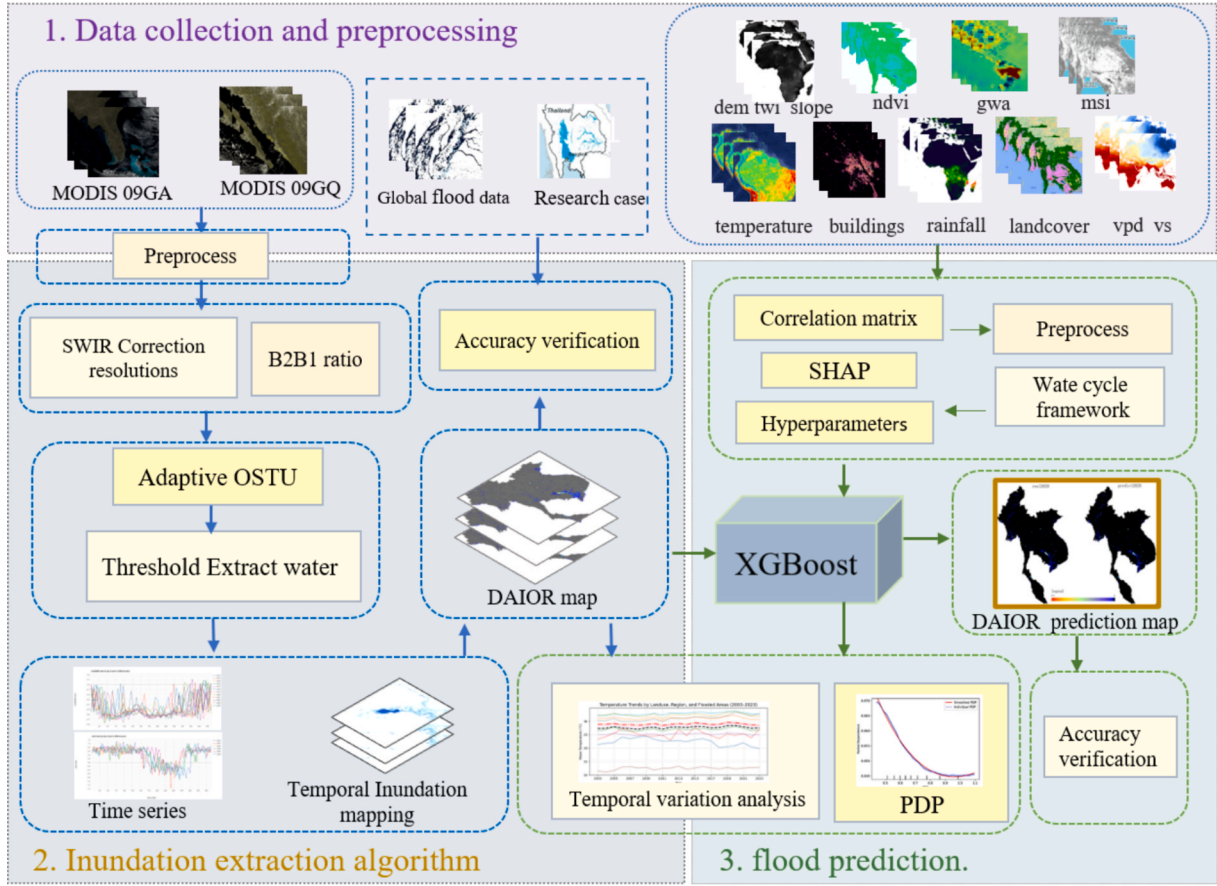


Fig. 2. Workflow overview. Stage 1 (purple) collected and preprocessed dataset of flood; Stage 2 (blue) long-time inundation extraction algorithm; Stage 3 (green) flood susceptibility prediction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

infrared (NIR) band (band 2; 841–876 nm) with 250 m resolution. MOD09GA is also adopted because it contains the 500-meter red band (band 1; 621–670 nm) shortwave infrared (SWIR) band (band 7; 1628–1652 nm), the red band is used to calibrate the MOD09GQ in the 500 m band. To enhance spatial resolution for water detection, the shortwave-infrared (SWIR) band (band7; 1628–1652 nm) commonly used for surface water identification, was sharpened to 250 m using a reflectance correction algorithm (Gumley et al., 2003; Tellman et al., 2021). Then the red band, NIR band and SWIR band with 250-m resolution in MODIS data were obtained. A threshold-based inundation algorithm was applied, utilizing an index called B1B2, defined as:

$$B1B2_{ratio} = \frac{NIR + 13.5}{red + 1081.1} \quad (1)$$

The constants 13.5 and 1081.1 used in the B1B2 ratio originate from the empirical calibration performed by Marsalek et al. (2006) using large global MODIS datasets. These offsets were introduced to compensate for sensor-related biases, atmospheric effects, background reflectance variability, and land-cover differences in both NIR and red bands. The small offset in the NIR band (13.5) stabilizes the low reflectance of water, whereas the larger offset in the red band (1081.1) corrects for systematic background bias, enabling a robust separation between water and non-water pixels across diverse regions. These constants are not physical parameters but empirically optimized coefficients designed to make the ratio globally applicable for flood detection. A pixel was classified as flooded if it satisfied all of the following conditions:

$$water = red < C \cap B1B2_{ratio} < K_1 \cap SWIR < K_2 \quad (2)$$

where C is the empirical threshold ($C = 2027$) obtained by the United

States Geological Survey (USGS) by regression to exclude non-surface within the red zone from river flow data ($R^2 = 0.91$) (Marsalek et al., 2006). K_1 is the threshold for determining the B1B2 ratio, while K_2 limits the upper bound of SWIR reflectance to suppress confusion with moist soils and other low-reflective materials, all of which are determined by the adaptive Otsu algorithm in this study.

3.1.1. Adaptive Otsu-based thresholding for flood detection

To improve detection performance in heterogeneous flood scenes, an adaptive version of the Otsu algorithm was developed to extract flood inundation areas by dynamically adjusting dual thresholds (K_1 and K_2) based on local spatial conditions.

The classical Otsu algorithm automatically determine an optimal threshold that best separates an image into two classes — foreground and background — based on the statistical distribution of pixel values. The method assumes that the histogram of the input image exhibits a bimodal pattern, where the two peaks correspond to water and non-water reflectance responses. The Otsu algorithm evaluates all possible threshold values and selects the one that maximizes the between-class variance, which quantifies how distinctly the two classes are separated. A higher between-class variance indicates a clearer discrimination between water and non-water pixels, resulting in more reliable segmentation (Otsu, 1975).

However, when water bodies occupy a small fraction of an image ($<30\%$), the spectral overlap between water and background often results in a unimodal histogram, making the Otsu method less effective (Li et al., 2019). To overcome this limitation, a spatially adaptive framework was proposed that automatically detects target subregions where bimodal distributions are more likely to occur, typically at the boundary of inundated areas.

The ICP was divided into 121 spatial blocks using a global grid system. Each block was initially filtered based on topographic criteria (elevation <300 m and slope <30°) to exclude high-relief regions less susceptible to flooding. Within each filtered block, a moving window approach (20 km × 20 km) was employed to generate 210–220 sub-regions for further analysis. The moving windows were implemented as a systematic sliding window scheme, where fixed-size windows were shifted across each block with a uniform spatial step, ensuring consistent spatial coverage rather than random sampling. Flood-prone periods were identified using rainfall time series to select representative MODIS imagery. After removing cloud-contaminated pixels, windows with more than 50% valid observations were retained. The Otsu algorithm was then applied to each valid window, and thresholds were accepted only if they fell within empirically defined ranges (e.g., $0.3 < B1B2 < 1.2$; $250 < SWIR < 1000$) (Tellman et al., 2021). These thresholds, representing locally optimal water–land separation values, were aggregated across all windows. The final global threshold for each image was computed as the average of all accepted local thresholds. All key parameters involved in the block division, sliding-window configuration, rainfall-based period selection, and threshold acceptance criteria are summarized in Table S1 in the supplementary materials to ensure reproducibility.

To enhance user accessibility and processing efficiency, we developed an interactive web-based application on GEE to support dynamic flood analysis. The platform features a GUI that enables users to define spatial and temporal parameters without coding. It automatically generates inundation occurrence maps and extracts time series of key indices such as SWIR reflectance and the B1B2 ratio. These time series are used to detect flood onset, duration, and peaks using adaptive thresholding. Color gradients from black (low) to red (high) visualize DAIOR (Fig. 3). This system has demonstrated high accuracy in capturing major events in Bangkok (e.g., Oct–Dec 2016) and is designed to support operational flood surveillance, particularly in data-scarce regions.

3.1.2. Inundation occurrence mapping and interactive monitoring platform

To support large-scale, dynamic flood monitoring, a flood event detection method based on times series based on the flood range of each image was extracted, and the inundation occurrence identified in the time series of each pixel was calculated by combining with formula 3.

$$f_w = \frac{N_w}{N_a} \quad (3)$$

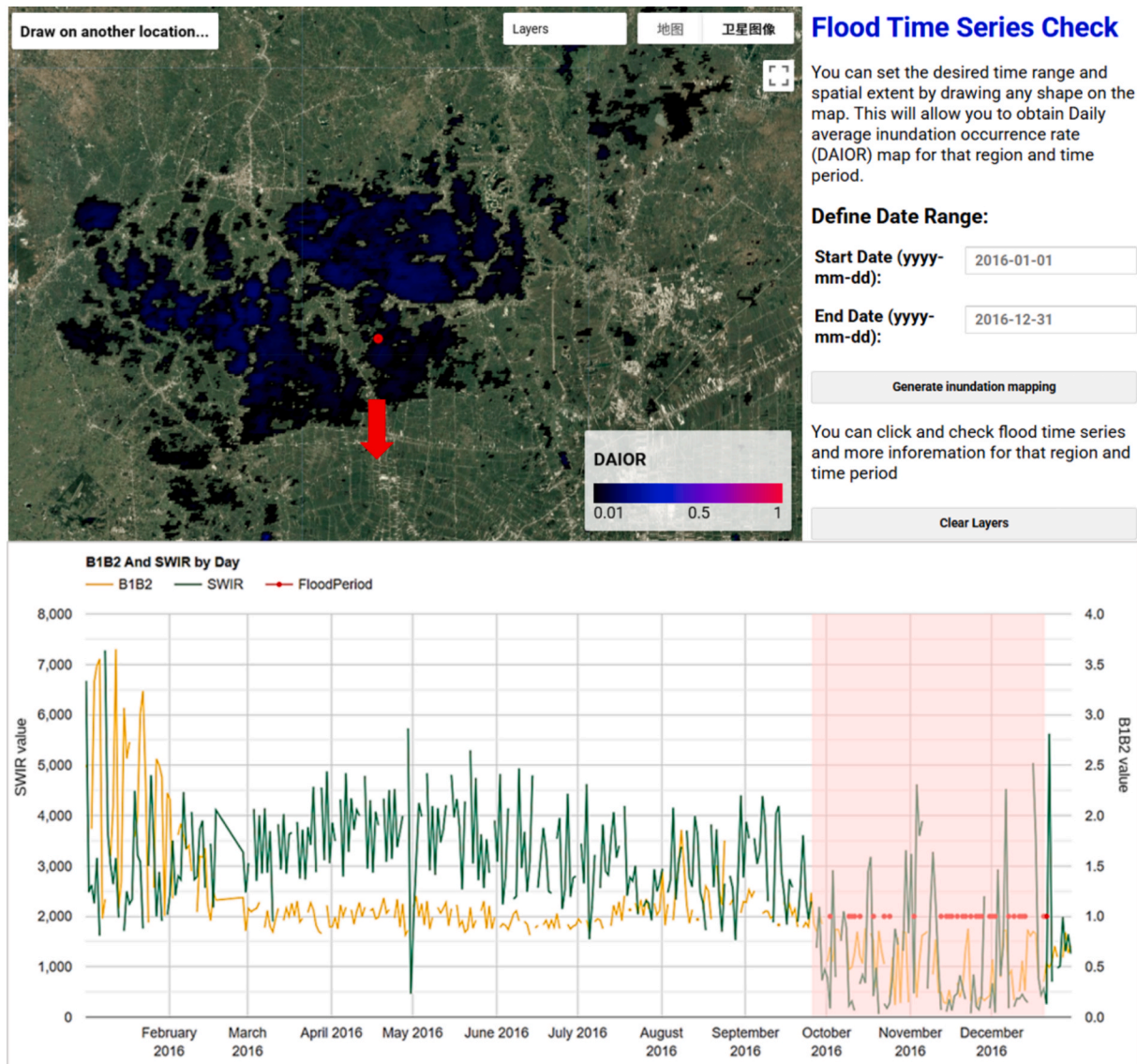


Fig. 3. Schematic diagram of the interface of the flood time series detection system and the flood time series.

where N_w represents the number of times a pixel was identification as water, and N_a refers to the total number of satellite image acquisitions (scenes) available during the analysis period. The f_w reflects how often a pixel is classified as water over a given period. Areas subject to prolonged or recurrent flooding exhibit higher inundation occurrence values.

Building on this concept, we define the DAIOR as the spatially explicit measure of inundation occurrence rate at each pixel, directly derived from f_w . In this study, DAIOR is calculated on an annual basis as the proportion of valid satellite observation in which a pixel is classified as inundated. It represents scene-based inundation occurrence rather than probabilistic flood frequency or return-period estimates. DAIOR provides a consistent framework to quantify how often each location is flooded, facilitating comparisons across space and time and serving as the response variable for subsequent analysis of flood drivers.

This study extracted the occurrence rate at which each cell is flooded during the year. In this process, no cloud masking, image compositing, or temporal reconstruction techniques were applied to the satellite imagery. This decision was made to maximize the temporal density and completeness of the time series data, thereby retaining all potentially useful observations, even in the presence of occasional cloud contamination. By preserving the full sequence of original scenes, the approach aims to capture more frequent and subtle flood signals that might otherwise be lost during preprocessing.

3.1.3. Flood extraction evaluation metrics

To assess the accuracy of flood extraction, a confusion matrix is used, which compares predicted classifications to satellite-based ground truth, to provide a detailed and intuitive summary of the model's performance (Ulmas and Liiv, 2020). It compares the predicted class labels with the true class labels, yielding four key metrics: True Positives or TP (correctly predicted positive instances), True Negatives or TN (correctly predicted negative instances), False Positives or FP (instances incorrectly predicted as positive), and False Negatives or FN (instances incorrectly predicted as negative). This matrix enables the calculation of various performance metrics, such as accuracy, precision, recall, and the F1 score, offering a more nuanced understanding of the model's effectiveness. From this, we computed the key evaluation metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

$$\text{F1 Score} = \frac{2(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (7)$$

These metrics provide a comprehensive evaluation of the algorithm's performance and robustness across diverse flooding scenarios.

To validate the inundation extraction results a comparative accuracy assessment is conducted using both standard thresholding and adaptive Otsu-based thresholding methods.

Tellman et al. (2021) conducted a sampling-based sensitivity analysis to determine how the number of validation points affects flood-map accuracy. They generated 500 stratified validation points for each of ten major flood events, and repeatedly subsampled 0–500 points without replacement. By analyzing the resulting variability in precision, recall, and overall accuracy, they demonstrated that accuracy metrics stabilize once ≥ 250 points are included. Following this concept, we selected 10 representative flood events for validation. For each event, stratified random sampling was applied to extract 500 reference points (with a 2:1 ratio of water to non-water pixels), and the results were compared against the DFO database.

3.2. Flood susceptibility prediction

To quantify flood susceptibility across the ICP, we developed a machine learning based framework integrating multi-source environmental variables and satellite derived flood observations. Predictors were structured hierarchically based on hydrological processes, encompassing atmospheric, surface, and subsurface drivers.

The workflow includes four components: (1) selection and pre-processing of features data to ensure consistency and reduce correlation; (2) flood susceptibility prediction modeling using XGBoost with Bayesian hyperparameter tuning; (3) interpretation via SHAP and PDPs to identify dominant and nonlinear driver effects; and (4) model evaluation using RMSE and R^2 . This approach enables spatially explicit, interpretable flood prediction and enhances understanding of underlying physical controls.

3.2.1. Selection and pre-processing of flood influencing data

Flood occurrence in the ICP results from complex interactions among meteorological, hydrological, and land surface processes. To ensure that the predictor set reflects these multi-component controls, this study adopts a macroscopic, water-cycle-oriented approach to investigate flood drivers.

Previous studies have shown that climate warming enhances the atmosphere's moisture-holding capacity, intensifying extreme precipitation events (Chagas et al., 2022; Donat et al., 2016; Fischer and Knutti, 2016; Trenberth, 2011). Land-surface modifications such as agricultural intensification, urban expansion and groundwater extraction alter infiltration, surface runoff and baseflow regimes, thereby modifying flood response (de Graaf et al., 2019; Zhang et al., 2018). In hydrologically complex regions such as the ICP, flood generation is also influenced by spatial rainfall heterogeneity (Sharma et al., 2018), temporal evolution and runoff of precipitation, but also on the soil moisture conditions (Blöschl et al., 2019; Hamlet and Lettenmaier, 2007).

Therefore, we compiled a comprehensive set of candidate predictors representing water input (precipitation), evaporative demand (temperature, wind speed, vapor pressure difference), land-surface retention and permeability (NDVI, land cover, built-up area, DEM, slope, TWI, landcover) and soil-groundwater moisture status (GWA, MSI). This ensures that the dataset captures variables relevant to infiltration, runoff generation, evapotranspiration and soil-water storage.

To ensure consistency among spatial predictors and reduce redundant information, we first examined pairwise correlation using Spearman's rho coefficients (Dormann et al., 2013). Variables with strong correlations ($|r| > 0.7$) were reviewed to avoid unnecessary duplication (Tehrany et al., 2019). We then computed Variance Inflation Factors (VIFs) to characterise the multivariate dependency structure inherent to hydroclimatic variables (O'Brien, 2007). Because strong coupling among temperature, VPD, groundwater anomalies, and other climate-driven indicators is physically inherent to large-scale water-cycle processes, VIFs were used diagnostically rather than as a basis for feature elimination. These dependencies are inherent to large-scale water-cycle processes — e.g., VPD is mathematically dependent on temperature, and groundwater anomalies co-vary with climatic and land-surface conditions. Moreover, the tree-based XGBoost model is inherently robust to correlated predictors, as it relies on recursive partitioning rather than coefficient estimation. Predictor importance was subsequently evaluated using SHAP analysis to ensure stable and interpretable model behavior. This procedure preserves the physical completeness of the water-cycle representation (atmospheric inputs, land-surface processes and subsurface conditions), while ensuring a stable and interpretable modelling framework.

To ensure the reliability and comparability of the input datasets, all remote sensing variables underwent a standardized preprocessing workflow (Amani et al., 2020). All datasets were spatially clipped to the study area and temporally filtered to the period of interest (Chander et al., 2009; Roy et al., 2010). To enable pixel-wise analysis, datasets

were resampled to a common spatial resolution of 250 m (Gao et al., 2006). Outliers were detected and removed using interquartile range filtering (Leys et al., 2013). Finally, continuous variables were standardized using z-score normalization to reduce scale-related bias and enhance model convergence (Kuhn and Johnson, 2013).

3.2.2. Modeling flood susceptibility prediction

XGBoost (Extreme Gradient Boosting) is a machine learning algorithm based on the gradient boosting framework, designed to improve the performance of traditional Gradient Boosting Decision Trees (GBDT) by enhancing and regularizing the loss function. The core idea of XGBoost is to iteratively improve the model's predictions by combining weak learners (typically regression trees) with weighted updates, which reduces bias and variance, thereby increasing prediction accuracy. Compared to traditional GBDT, XGBoost improves upon it by using a second-order Taylor expansion to approximate the loss function, allowing for more robust updates and better resistance to overfitting. XGBoost is not only suitable for regression and classification tasks but also capable of handling large-scale datasets. In flood prediction, XGBoost's advantages lie in its ability to handle complex spatiotemporal data and nonlinear relationships. Since floods are influenced by various factors such as rainfall, topography, and land use, XGBoost can capture complex interactions between these features, thereby improving the accuracy of flood predictions.

In this study, DAIOR was used as the regression target variable. In total, 187,239,506 samples (93,519,753 pixels $\times$ 3 years) were used and randomly split into training and testing sets at a ratio of 9:1 using a fixed random seed to ensure reproducibility (Belgiu and Drăguț, 2016). To examine the impact of temporal context on model performance, three temporal configurations — one-year, two-year, and three-year datasets — were assessed. Specifically, 18 experimental combinations were constructed: five with one-year inputs, four with two-year inputs, and three with three-year inputs (Maxwell et al., 2018).

To improve model robustness and avoid overfitting, Bayesian optimization was used to tune XGBoost hyperparameters (Chen, 2016; Snoek et al., 2012). The objective function was initialized as a black box, and the optimizer was run with 5 random points and executed for 25 iterations. A fixed random seed was used to ensure reproducibility and detailed logs were maintained to facilitate traceability. This process helped identify optimal configurations efficiently and enhanced model generalization. Despite the large dataset size, each training cycle was completed in under one minute on a high-performance workstation (NVIDIA RTX 3090 GPU), highlighting the computational efficiency of the approach.

For consistency across experiments, 2020 was used as the representative prediction year when summarizing the temporal-configuration results. The year 2020 is centrally located within the 2015–2023 window during which all predictor variables maintain stable remote-sensing availability and uniform spatial coverage. To verify that this choice does not introduce bias, the same models were also applied to predict condition in 2021.

3.2.3. Model interpretation: SHAP and PDPs

To enhance model interpretability and explore the relative influence of individual flood-driving factors, linking data-driven learning with conceptual hydrological understanding, we employed two model explanation techniques: SHapley Additive exPlanations (SHAP) and partial dependence plots (PDPs).

The SHAP framework was employed to interpret the XGBoost model by quantifying the contribution of each predictor to individual predictions (Kim and Kim, 2022). Specifically, we used the TreeSHAP algorithm, which is designed for tree-based ensemble models and allows exact and efficient computation of SHAP values for XGBoost (Lundberg et al., 2020; Lundberg and Lee, 2017).

SHAP values represent the marginal contribution of each predictor to the deviation of a model prediction from the baseline expectation,

averaged over all possible feature coalitions. By aggregating SHAP values across all pixels, we derived a global importance ranking of flood drivers, enabling transparent identification of dominant controls such as NDVI, land use, temperature, and groundwater anomalies (Kaiser et al., 2022).

PDPs are used to visualize the marginal effect of one or more features on the predicted outcome of a machine learning model, while averaging out the influence of all other variables (Friedman, 2001). In this study, PDPs were employed to interpret the impact of key climate and land surface variables — such as NDVI, land use, and groundwater anomalies — on inundation occurrence predictions generated by the XGBoost model. By isolating the relationship between each variable and the model output, PDPs help to uncover non-linear patterns and potential thresholds that contribute to flood risk. This interpretability enhances the physical plausibility of the model results and provides insights into the dominant flood-driving mechanisms. In this study, PDPs were applied to key climatic and land surface features, revealing how changes in these variables influence inundation occurrence across different ranges.

3.2.4. Evaluation methodology for flood prediction models

To assess the accuracy and generalization capability of the flood susceptibility prediction model, we employed two widely used regression evaluation metrics: Root Mean Square Error (RMSE) and the coefficient of determination (R^2), which are commonly adopted in hydrological and environmental modeling. RMSE is defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (8)$$

where y_i is the observed flood susceptibility, $\hat{y}_i$ is the predicted value, and n is the number of observations. RMSE reflects the average magnitude of prediction errors, lower values indicate higher predictive accuracy and reduced bias (Willmott and Matsuura, 2005). R^2 is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (9)$$

where $\bar{y}$ is the mean of observed values. R^2 represents the proportion of variance in the observed data explained by the model and reflects the overall goodness-of-fit. A value closer to 1 indicates stronger model performance.

To demonstrate the superiority of the proposed XGBoost based models, the performance of several mainstream ML and DL models RF, SVR, LightGBM, LSTM, and CNN was compared using data from 2018. All models were trained and tested on the same preprocessed dataset under comparable parameter optimization settings to ensure fairness. In addition to R^2 and RMSE, mean absolute error (MAE), time, and memory usage are added for comparison (Corona and Hogue, 2025; Nash and Sutcliffe, 1970; Shen et al., 2018). MAE is defined as:

$$MAE = \frac{1}{n \sum_{i=1}^n |y_i - \hat{y}_i|} \quad (10)$$

where y_i is the observed inundation occurrence rate, $\hat{y}_i$ is the predicted value, and n is the number of observations. MAE represents the average magnitude of errors between predicted and observed values, without considering their direction. It reflects the model's ability to capture deviations in absolute terms, where a smaller MAE indicates higher prediction accuracy.

4. Results

4.1. Satellite-observed inundation

A total of 21 annual inundation occurrence maps were generated

using all available MODIS imagery from 2003 to 2023 across the study area, with one map per year (Fig. 4). These maps illustrate the inter-annual variability of inundation in the Tonle Sap River Basin and broader ICP. Fig. 5 summarizes the annual extent of inundation as a percentage of the total study area. From 2003 to 2023, the proportion of the flooded area exhibited fluctuations, but the average annual flood coverage remained relatively stable at approximately 5%. The proportion of severely inundated areas — defined as regions flooded persistently over two weeks within a year — also remained steady at around 1–2%. In terms of flood impact area, the area affected was the largest in 2011 (5.4% in the ICP), followed by 2007 (4.8%), 2004 (4.75%), 2006 (4.6%), 2015 (3.51%), followed by 2020 (3.61%), and 2019 (3.73%).

The duration of flooding also varied across years. In 2011, 2.16% of the study area experienced flood conditions lasting longer than two

weeks — the highest in the two-decade period — followed by 2006 (1.72%), 2004 (1.53%), and 2015–2016 (0.97%). After 2019, flood extent and duration showed minimal interannual variation, suggesting a recent stabilization in flood dynamics. The extreme floods of 2011, driven by enhanced monsoonal activity and typhoon events across Southeast Asia, caused widespread inundation in Thailand, severely impacting agricultural and industrial zones, including the Bangkok metropolitan area. In contrast, the flood anomalies observed in 2015 occurred due to a strong El Niño phenomenon, which caused widespread drought around the world.

To analyze spatial patterns of inundation, inundation occurrence rate values were classified using the natural breaks (Jenks) method into five categories: lowest (<2%), lower (2–7%), medium (7–14.4%), higher (14.4–24.2%), and highest (24.2–38.6%). Fig. 6 illustrates the spatial

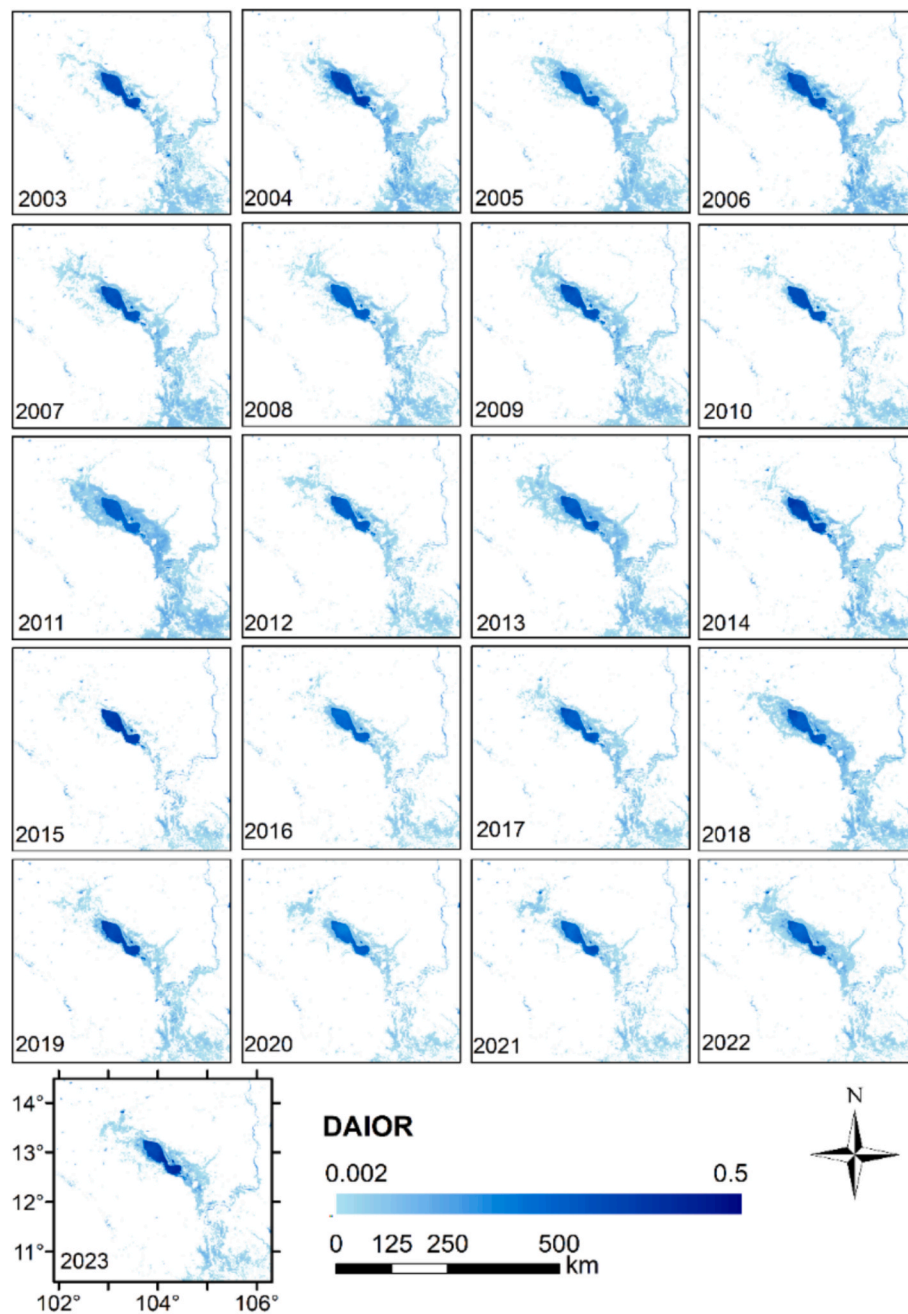


Fig. 4. DAIOR of floods between 2003 and 2023 (in the case of the Saridong Lake basin). The values range from 0.002 to 0.5, visualized in a color gradient from light blue (low) to dark blue (high), indicating the probability of each pixel being inundated in a given year. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

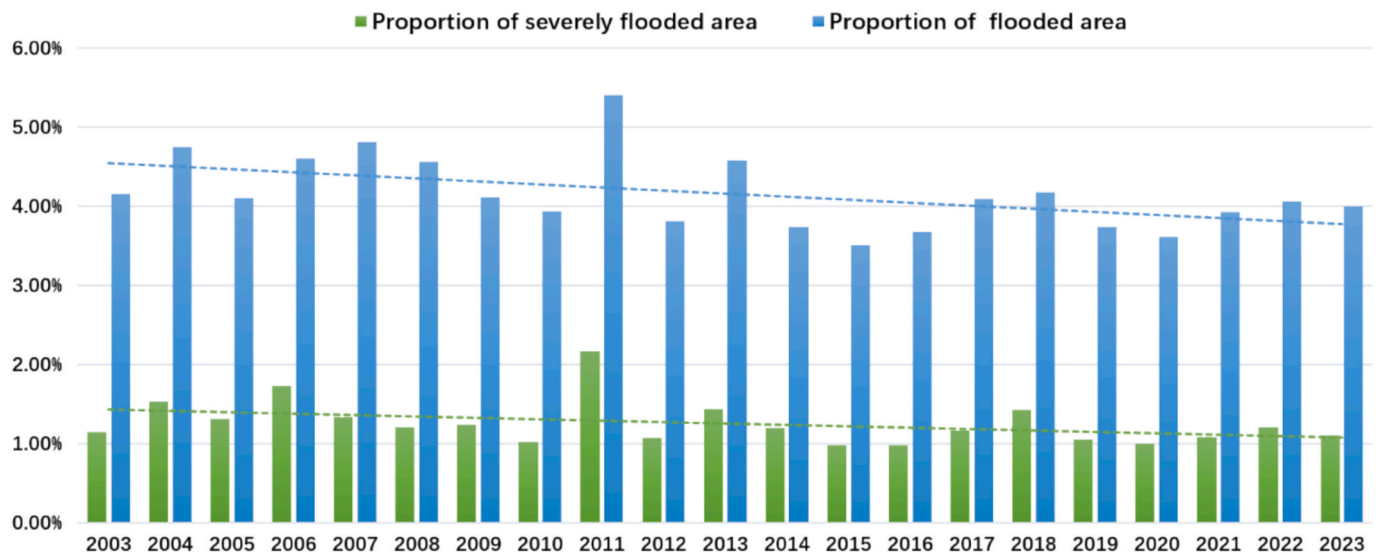


Fig. 5. The total flooded area (blue) and severely flooded area (green) are shown as a percentage of the total study area. The dotted line shows the trend. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

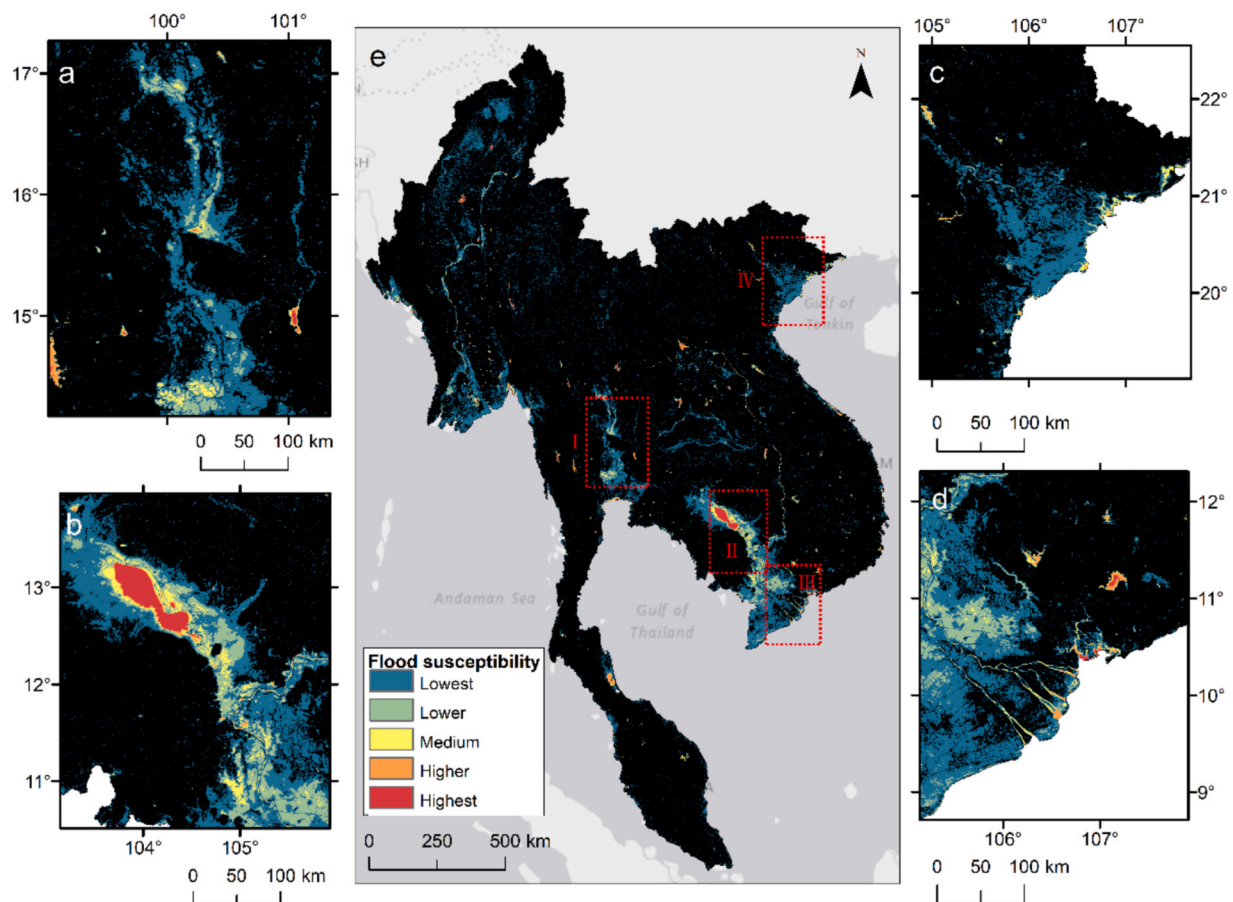


Fig. 6. Spatial distribution of flood susceptibility in ICP (2003–2023). The central main map shows the overall spatial pattern of flood susceptibility in ICP from 2003 to 2023, and the color indicates the susceptibility level, from the lowest (blue) to the highest (red). a-d highlight the typical high-risk areas in (a) the middle Mekong River, (b) central Thailand, (c) the Red River Delta, and (d) the Mekong Delta. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

distribution of high-susceptibility flood zones over the 20-year period. Areas with the highest flood frequencies are primarily concentrated along major rivers — including the Mekong, Red, and Salween Rivers — and in low-lying deltaic regions such as the Mekong and Red River

Deltas. Highland and mountainous regions, such as northern Thailand, display markedly lower flood frequencies. Specific spatial patterns are illustrated in selected subregions:

- Fig. 6a highlights the flood-prone zone along the middle reaches of the Mekong River near the Laos – Thailand border. The strip-like distribution suggests possible accumulation on alluvial plains downstream of mountainous catchments.
- Fig. 6b shows widespread flood-prone zones in the central plains of Thailand, particularly north of Bangkok. This area is characterized by low elevation, dense hydrological networks, and extensive agricultural development within the Chao Phraya River Basin.
- Fig. 6c depicts flood-prone coastal zones in the Red River Delta of northern Vietnam, where flood susceptibility is amplified by the interaction of tidal dynamics and heavy rainfall.
- Fig. 6d focuses on the Mekong Delta in southern Vietnam, a region with a dense network of interwoven rivers. Flood levels exhibit a centripetal gradient — highest near river confluences and gradually decreasing outward — indicative of strong hydrodynamic control.

The 2003–2023 flood susceptibility analysis provides a spatially explicit, 21 years record of inundation dynamics across the region. Of the total inundated area (39,003 km²), approximately 3,730 km² (0.18%) experienced inundation more than 4.8% of the time — equivalent to over two weeks annually — while 23,166 km² (1.12%) were flooded at least once per year. These flood susceptibility maps can help inform urban planning and agricultural decision-making by illustrating inundation occurrence and highlighting spatiotemporal inundation trends that are not visible on interannual time scales.

As expected, zones of high inundation susceptibility in the ICP are predominantly concentrated along river corridors, floodplains, and lake margins, where hydrodynamic fluctuations driven by monsoonal precipitation result in seasonal or quasi-permanent water presence.

As summarized in Table 2, the Inundation Occurrence Dataset (this study) achieved a slightly higher average overall accuracy (0.935) than the Global Flood Database (GFD) (0.923). Marginal improvements were also observed in precision (0.930 vs. 0.920), recall (0.960 vs. 0.950), and F1-score (0.942 vs. 0.939). These results indicate that combining regionally adaptive thresholding with a continuous MODIS time series yields modest but consistent improvements in flood-extent mapping accuracy, particularly in complex or heterogeneous floodplain environments where spatial and temporal variability is high.

Uncertainty associated with inundation extraction and model interpretation should be considered when evaluating these results. First, the 250 m resolution of MODIS limits the detection of small or narrow water bodies, which may remain unresolved or misclassified (Chini et al., 2019). Mixed pixels containing both water and sediment-rich surfaces can further introduce classification ambiguity due to spectral similarity with land. In forested areas, dense canopy cover limits optical visibility of underlying floodwaters, resulting in systematic underestimation of inundation extent (Hawker et al., 2020). Additionally, MODIS provides a maximum of two observations per day, which may be insufficient to capture short-duration inundate pulses, particularly flash floods. These factors contribute to uncertainty in reported inundation occurrence patterns and should be considered when interpreting spatial variation in DAIOR.

Threshold selection represents an additional source of uncertainty. The adaptive Otsu-based approach employed here derives local threshold from spatial reflectance statistics within moving windows.

Table 2

Performance comparison of our inundation occurrence dataset and the Global Flood Database's flood range extraction based on precision, recall, F1 score, and overall accuracy.

	Precision	Recall	F1 Score	Overall Accuracy
Inundation Occurrence Dataset (this study)	0.930	0.960	0.942	0.935
Global Flood Database (GFD)	0.920	0.950	0.939	0.923

Empirical results indicate that the optimal threshold value is inversely related to the proportion of inundated pixels within a window, such that larger the inundated area yield lower thresholds. Although a final threshold is computed by averaging of all accepted local thresholds, residual variability persists due to landscape heterogeneity and scene-specific conditions. This uncertainty primarily affects the magnitude of inundation occurrence estimates rather than the relative spatial patterns identified in the results.

4.2. Flood susceptibility prediction and model validation

The predictive capability of the XGBoost model was evaluated using multiple feature interpretation techniques and quantitative performance metrics. Fig. 7 compares observed flood DAOIR (real 2020) and model-predicted flood susceptibility (predict 2020) for 2020 across two representative areas. The predicted maps closely reproduce the spatial distribution of high-risk zones, particularly along major rivers and adjacent floodplains. The results demonstrate that the model effectively captures spatial patterns of flood-prone areas, reinforcing its suitability for flood risk mapping.

To ensure input data integrity, inter-variable relationships were examined using Spearman's rho correlation coefficients. As shown in Fig. 8, all pairwise correlations among the 12 controlling variables were below the commonly accepted threshold of 0.7 (Tehrany et al., 2019), indicating the absence of strong pairwise associations. This suggests the statistical independence of the features and ensures model robustness.

Model interpretability was assessed using SHAP values. The SHAP summary plot (Fig. 9) and feature importance analysis (Fig. 10) highlights NDVI as the most influential variable in flood susceptibility prediction. It's wide range of SHAP values indicates a strong nonlinear effect on model output. NDVI and temperature were also found to be significant contributors, followed by land use. These variables directly affect hydrological processes such as infiltration, runoff generation, and evapotranspiration. In contrast, features such as building density and TWI exhibited relatively limited influence. The results are consistent with hydrological theory: dense vegetation improves water retention, land use regulates surface permeability and drainage patterns, and terrain modulates flow accumulation. The agreement between model inference and physical hydrology supports the scientific validity of the model outputs.

Fig. 11 presents model validation results for the year 2020, using the best results case as an example. The scatter plot comparing predicted and observed DAIOR values shows strong alignment, with residuals closely clustered around the 1:1 line. The residual distribution approximates normality with low skewness, indicating minimal systematic bias. Predictions in high-risk zones (DAIOR >0.3) were generally unbiased, although some right-skewed deviations suggest room for improvement in treatment of high-risk conditions in data-driven flood modeling frameworks. Model performance improves with longer training periods.

As shown in Table 3, R² increases from 0.75 (one-year) to 0.82 (three-year), while RMSE remains low (0.02–0.04), indicating stable and accurate predictions for 2020. Across all one-year, two-year, and three-year input configurations, the resulting performance metrics (R² = 0.75–0.82; RMSE = 0.02–0.04) were consistent with those obtained for 2021, indicating stable predictive behaviour across years. Therefore, using 2020 as an illustrative example provides a clear and representative summary of model performance.

4.3. Analysis of factors affecting flood

Although precipitation is a necessary precursor to flooding, our results indicate that spatial variations in inundation occurrence are not primarily controlled by precipitation variability alone across the ICP. Instead, flood occurrence patterns are more strongly conditioned by interactions among topographic features, antecedent soil moisture, and

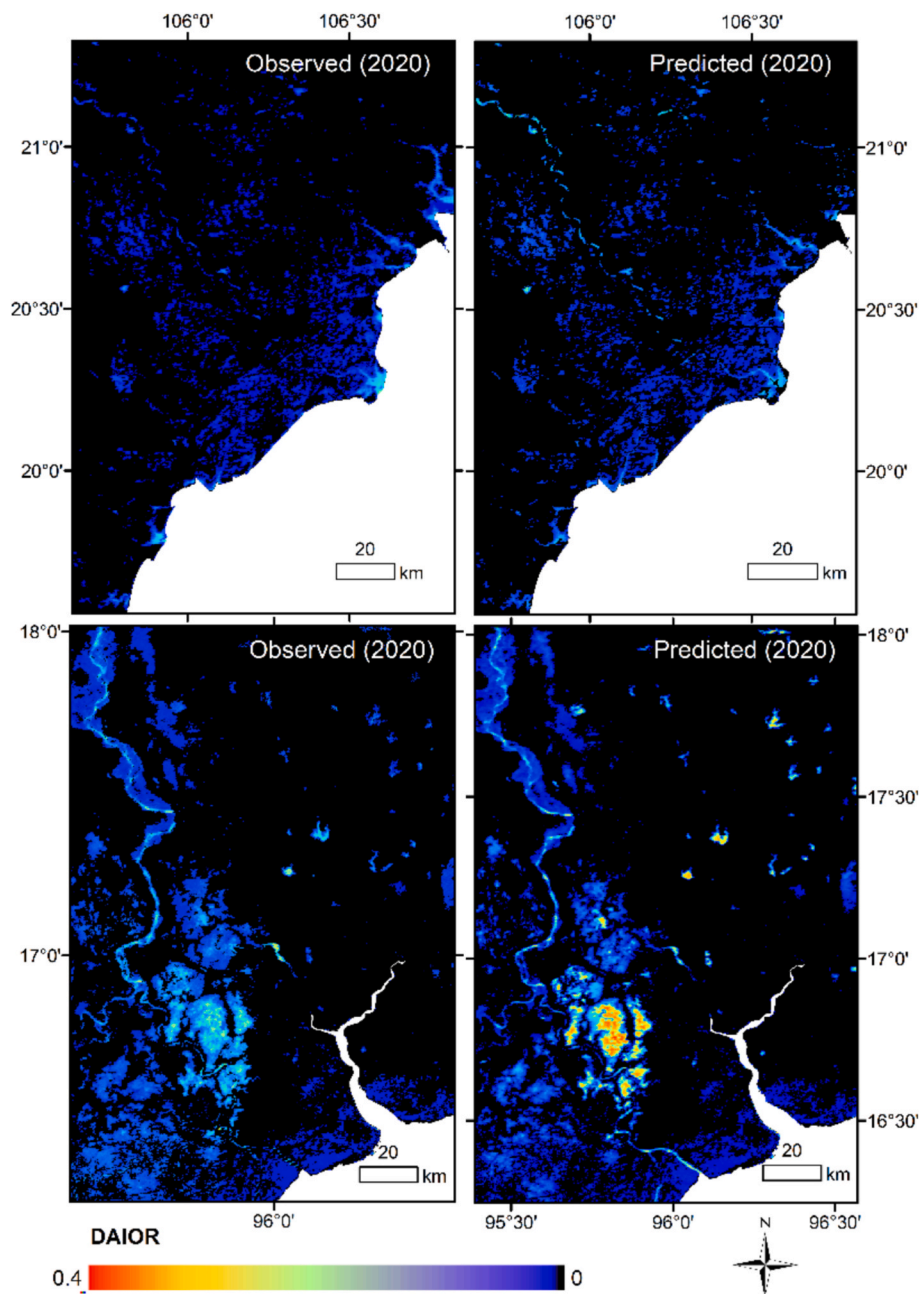


Fig. 7. Comparison of actual and forecasted flood susceptibility. Left is spatial distribution of observed inundation occurrence in 2020, and right is the predicted map. The legend indicates the level of DAIOR, color spectrum scales from dark blue (low) to red (high), with higher values indicating that the area is more prone to flooding. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

land-surface characteristics, which collectively regulate the landscape's sensitivity to rainfall. Despite relatively low daily rainfall (<8 mm) across much of Myanmar, surface flooding remains prevalent. This phenomenon is primarily attributed to high antecedent soil moisture, flat terrain, and extensive cropland coverage. As illustrated by the PDPs in Fig. 12, the influence of precipitation on flood risk is nonlinear and threshold-dependent.

Several environmental variables, particularly land use type and the TWI, also exhibit nonlinear, threshold-like relationships with flood occurrence rate. These responses highlight the significance of terrain-mediated hydrological processes. Elevation similarly demonstrates a nonlinear marginal effect, reinforcing the importance of spatial heterogeneity and coupled physical processes in flood dynamics.

The shaded blue confidence intervals in Fig. 12 offer insights into model uncertainty. Wider intervals, notably in precipitation and

building coverage, reflect higher predictive variance, which may arise from sparse sampling or local heterogeneity. For example, the staircase pattern observed in the PDPs for building coverage (Fig. 12b) likely stems from underrepresentation of urban structures in the training data, leading to discretized outputs.

The partial dependence plots (Fig. 12) demonstrate that the XGBoost model captures clear nonlinear and threshold-dependent relationships between key hydroclimatic and land-surface drivers and inundation occurrence. NDVI exhibits a strong and persistent negative association with DAIOR, with predicted values decreasing from approximately 0.10 to 0.02 as NDVI increases from ~ 0.1 to ~ 0.65 , corresponding to a marginal reduction on the order of 0.07–0.08. The most pronounced decline occurs when NDVI exceeds ~ 0.4 , indicating a substantial vegetation buffering effect on surface inundation. MSI shows a distinct threshold behavior near $MSI \approx 0.9$, below which inundation occurrence

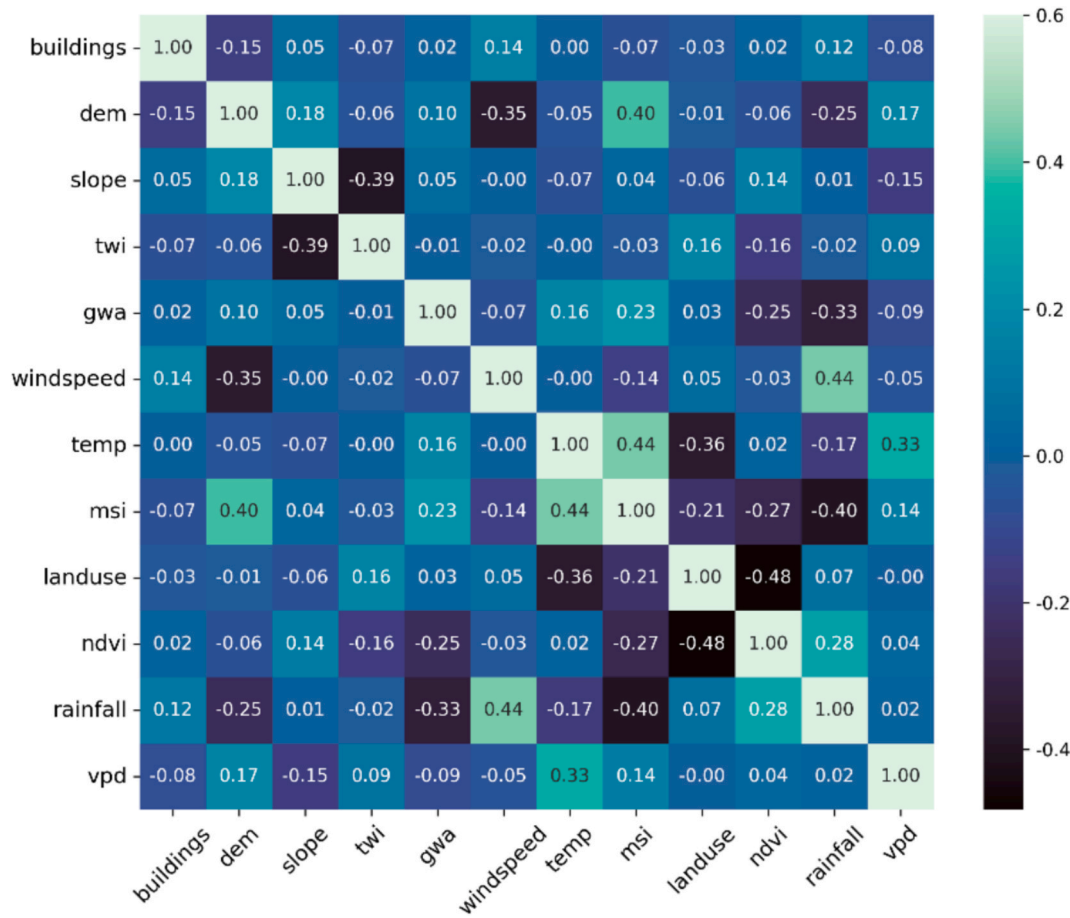


Fig. 8. Correlation matrix showing pairwise correlations among the 12 driving factors using Spearman's rho coefficients.

decreases rapidly by approximately 0.02–0.025, followed by a plateau and slight rebound at higher values. In contrast, precipitation displays a comparatively weak marginal effect, with total variation in DAIOR remaining below ~ 0.005 across the observed range, suggesting that rainfall intensity alone does not explain spatial differences in inundation occurrence. These nonlinear responses are consistent with established hydrological understanding of infiltration capacity, antecedent moisture conditions, and land-surface controls on runoff generation (Mosavi et al., 2018; Nearing et al., 2021; Shen, 2018).

All predictors in this study are applied at the pixel level (250 m resolution), enabling spatially explicit mapping of flood susceptibility across heterogeneous landscapes. Unlike traditional gauge-based approaches, which represent hydrological conditions at discrete point locations, this framework resolves fine-scale spatial variability in vegetation, topography, and soil moisture, thereby supporting more detailed interpretation of how localized land-surface changes influence flood risk (Brocca et al., 2017; Verger et al., 2014).

Among the predictors, NDVI, land use, and temperature emerged as the top three most influential factors. To further explore these effects, we examined their interactions. We analysed how land use and temperature interact (Figs. 13 and 14). Flood-prone areas are mostly located in warm zones (e.g., tropical monsoon regions above 26 °C). At 26–28 °C, cropland and urban areas show higher flood likelihood, but DAIOR declines when temperatures exceed ~ 30 °C — indicating a nonlinear “heat suppression” effect.

This pattern reflects a dual mechanism: spatially, warmer regions are more flood-prone due to flat terrain, intensive land use, and strong hydrological cycles. Temporally, extreme heat can reduce flood risk by drying out soils, especially during dry seasons. This highlights the importance of considering both climate background and dynamic

surface – climate interactions in flood models.

Land use significantly modulates both the magnitude and variability of flood exposure across the landscape. Among all land-use categories, cereal cropland consistently accounts for the largest inundated area ($>50,000$ km² annually) because it is predominantly located across low-lying, hydrologically connected flood plains that are naturally prone to inundation. The observed association therefore reflects landuse allocation to flood-prone terrain, rather than a causal contribution of cereal cropland to flooding. Despite its spatial dominance, cropland exhibits relatively stable flood extents over time, while more dynamic changes are observed in forested categories (Fig. 15b). For instance, evergreen needleleaf forest and deciduous needleleaf forest show higher interannual variability in DAIOR, possibly due to combined influences of canopy interception, soil moisture retention, and land management shifts.

Temporal changes in the total area of land use areas (Fig. 15c) provide additional context for interpreting these flood exposure patterns. While cropland area has slightly increased post-2010, shrubland and grassland areas have declined, partially offset by forest regeneration in some regions. This implies that land use change has not merely altered flood-prone surfaces, but also reshaped the underlying hydrological response patterns. Forested areas, especially needleleaf types, exhibit comparatively low flood frequencies and minor fluctuations in flooded area, consistent with their buffering role in flood mitigation. However, their declining extent implies reduced multi-years ecosystem service capacity.

Notably, although wetlands occupying only a small fraction of the total landscape (Fig. 15c), they exhibit the highest average inundation occurrence rate (Fig. 15b), underscoring a disproportionate hydrological burden per unit area. This pattern likely signals the degradation of natural buffering functions or disruptions in upstream catchment

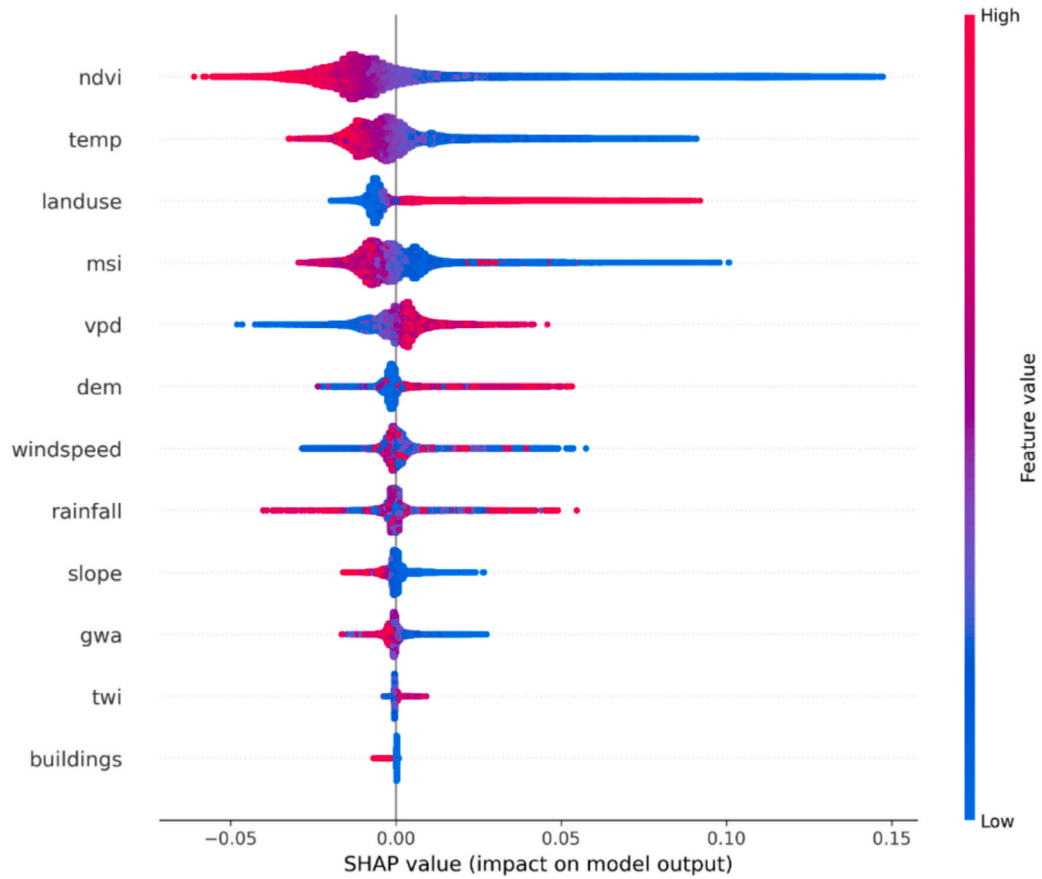


Fig. 9. SHAP summary plot illustrating the contribution and variability of each feature to model predictions.

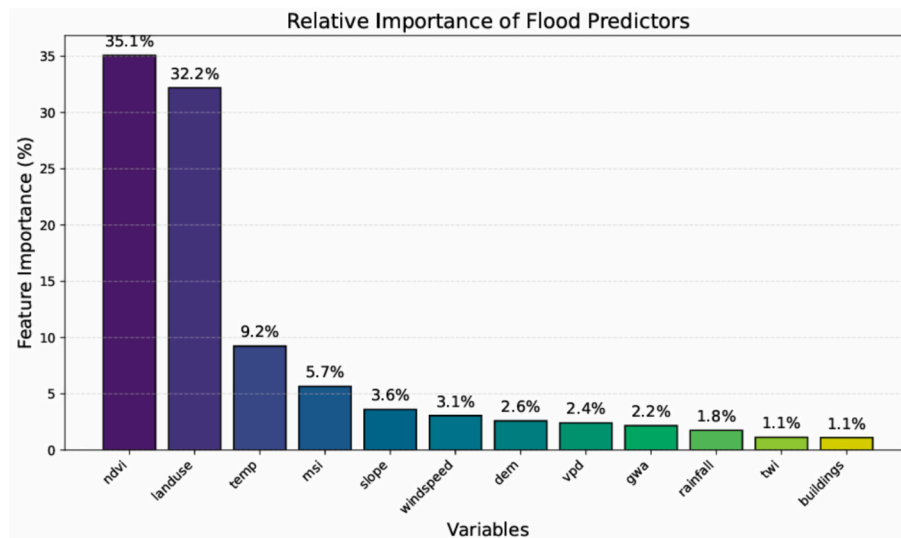


Fig. 10. Feature importance ranked by SHAP values.

connectivity. Similarly, urban areas, while spatially limited, show a notable post-2020 rise in inundation occurrence rate, peaking sharply in 2021. This may reflect the cumulative impact of increased impervious surfaces, drainage alterations, and climate-driven extremes, findings that align with other Southeast Asian cities undergoing rapid urban expansion.

In 2011, inundation occurrence rate spikes are particularly evident in cropland, shrubland, and forested areas, coinciding with one of the

most severe monsoonal flood events in Southeast Asia amplified by typhoon-driven rainfall and prolonged runoff accumulation. Conversely, 2015 — a strong El Niño year — exhibited unexpected increases in inundation occurrence across both urban and forested areas. Strong El Niño conditions are known to induce rainfall redistribution (Braud et al., 2013), characterized by a shift toward fewer rainfall days but higher event intensities (Cai et al., 2014; Power et al., 2013). Such high-intensity rainfall events, particularly when preceded by prolonged

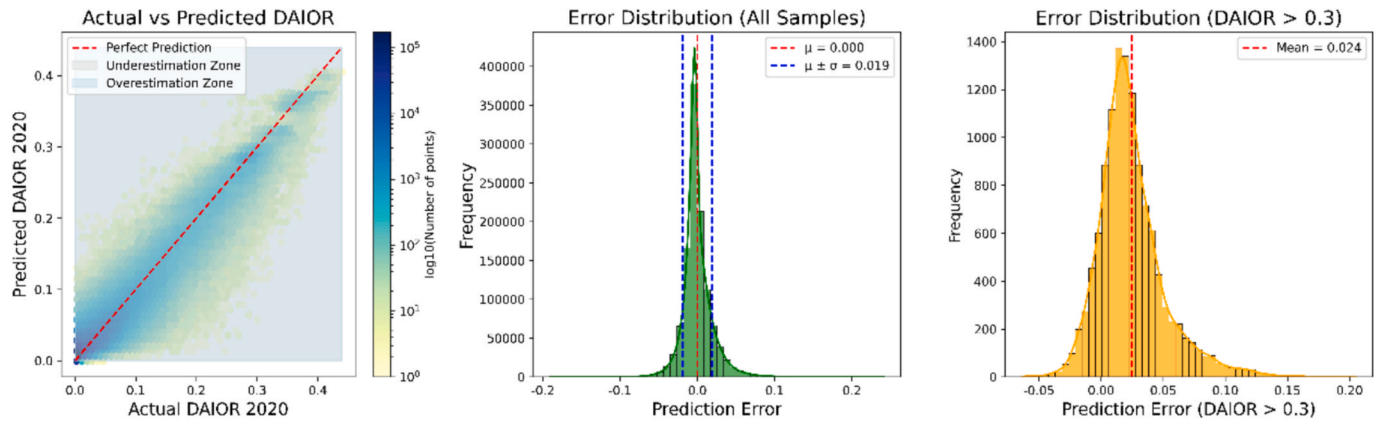


Fig. 11. Compares observed and predicted flood susceptibility in 2020. The left panel displays the relationship between actual and predicted inundation occurrence for 2020, with the red dashed line (perfect predictions) and over-underestimation zones. The middle panel illustrates overall prediction errors. The right panel highlights errors for extreme events (DAIOR > 0.3). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3

Validation results of XGBoost model under different training configurations for predicting DAIOR in 2020 and 2021. The table demonstrates stable model performance across prediction years and improved accuracy with longer training periods.

	Training Year	Predicted Year	R ²	RMSE
One-year	2015	2020	0.76	0.03
	2016	2020	0.75	0.03
	2017	2020	0.75	0.04
	2018	2020	0.77	0.03
	2019	2020	0.80	0.03
	2018	2021	0.76	0.04
	2019	2021	0.78	0.03
Two-year	2020	2021	0.80	0.03
	2015, 2016	2020	0.78	0.03
	2016, 2017	2020	0.79	0.03
	2017, 2018	2020	0.79	0.03
	2018, 2019	2020	0.80	0.03
Three-year	2018, 2019	2021	0.79	0.03
	2019, 2020	2021	0.80	0.03
	2015, 2016, 2017	2020	0.79	0.03
	2016, 2017, 2018	2020	0.80	0.03
	2017, 2018, 2019	2020	0.82	0.02
	2018, 2019, 2020	2021	0.81	0.02

dry periods, can substantially reduce effective soil infiltration capacity due to soil hydrophobicity, surface sealing, and degraded soil structure (Burch et al., 1989; Szejis et al., 2024). These processes enhance rapid surface runoff generation, increasing flood occurrence even in landscapes that are typically more absorptive. The sharp uptick in inundation occurrence rate across urban and wetland areas aligns with documented of extreme precipitation events in the Lower Mekong Basin in 2021 (Mekong River Commission, 2021), with flood impacts further amplified by expanding impervious surfaces and altered drainage dynamics that reduce infiltration and increase surface runoff (Gao et al., 2023). These observed spatiotemporal flood anomalies are consistent with key drivers identified in our XGBoost based variable importance analysis, which highlights cropland extent, slope, and urbanization intensity as dominant predictors of flood exposure. Together, these trends underscoring the need for adaptive land management, wetland restoration, and nature-based solutions to address increasing flood risks under a changing climate.

5. Discussion

5.1. 21-year remote sensing-derived flood DAIOR data

Recent studies emphasize the persistent scarcity of globally consistent global flood datasets with long temporal coverage, particularly those capable of capturing both extreme events and recurrent inundation dynamics at regional to continental scales (Misra et al., 2025). This study presents an inundation occurrence dataset derived from over two decades (2003–2023) of daily MODIS satellite imagery across the ICP. This dataset provides a temporally continuous, spatially extensive, and physically consistent representation of flood occurrence, enabling more robust analysis of spatiotemporal patterns and trends in flooding and suitable as a direct modeling target.

Compared to conventional event-based flood event databases such as the Global Flood Database (GFD) and the Dartmouth Flood Observatory (DFO), our results provide a more comprehensive spatiotemporal record by capturing all observable flood events over two decades. For example, in 2014, only two events were recorded in the international flood database, whereas our analysis detected extensive flooding along the southern shore of Tonle Sap Lake in Cambodia, corroborated by independent reports of casualties. Possible omissions in existing databases are due to information extraction based on discontinuous flood events and sparse observatories.

Although event-based flood products provide spatially complete footprints for each documented flood event, they archive floods as discrete occurrences rather than as a continuous time series. As a result, the temporal dimension is fragmented, with large gaps between events and limited representation of moderate, or recurrent inundation episodes. When aggregated over a year or multi-year period, these isolated event footprints do not form a temporally continuous record of flooding. Hydrological discharge datasets provide continuous temporal records, but they usually represent point measurements at gauging stations and therefore lack spatial continuity or the ability to depict full inundation patterns. Our MODIS-derived inundation occurrence dataset incorporates all available daily observations from 2003 to 2023, providing a continuous spatiotemporal record that captures both major flood events and recurrent or low-magnitude inundation dynamics that are typically underrepresented in event-based archives.

By leveraging daily MODIS imagery and the DAIOR metric, our study addresses these limitations and contributes to a more robust, multi-decadal representation of surface water dynamics. These inundation occurrence rate maps reveal otherwise undetectable patterns of seasonal and episodic inundation, offers a more complete representation of flood occurrence and supports improved water resource management, urban

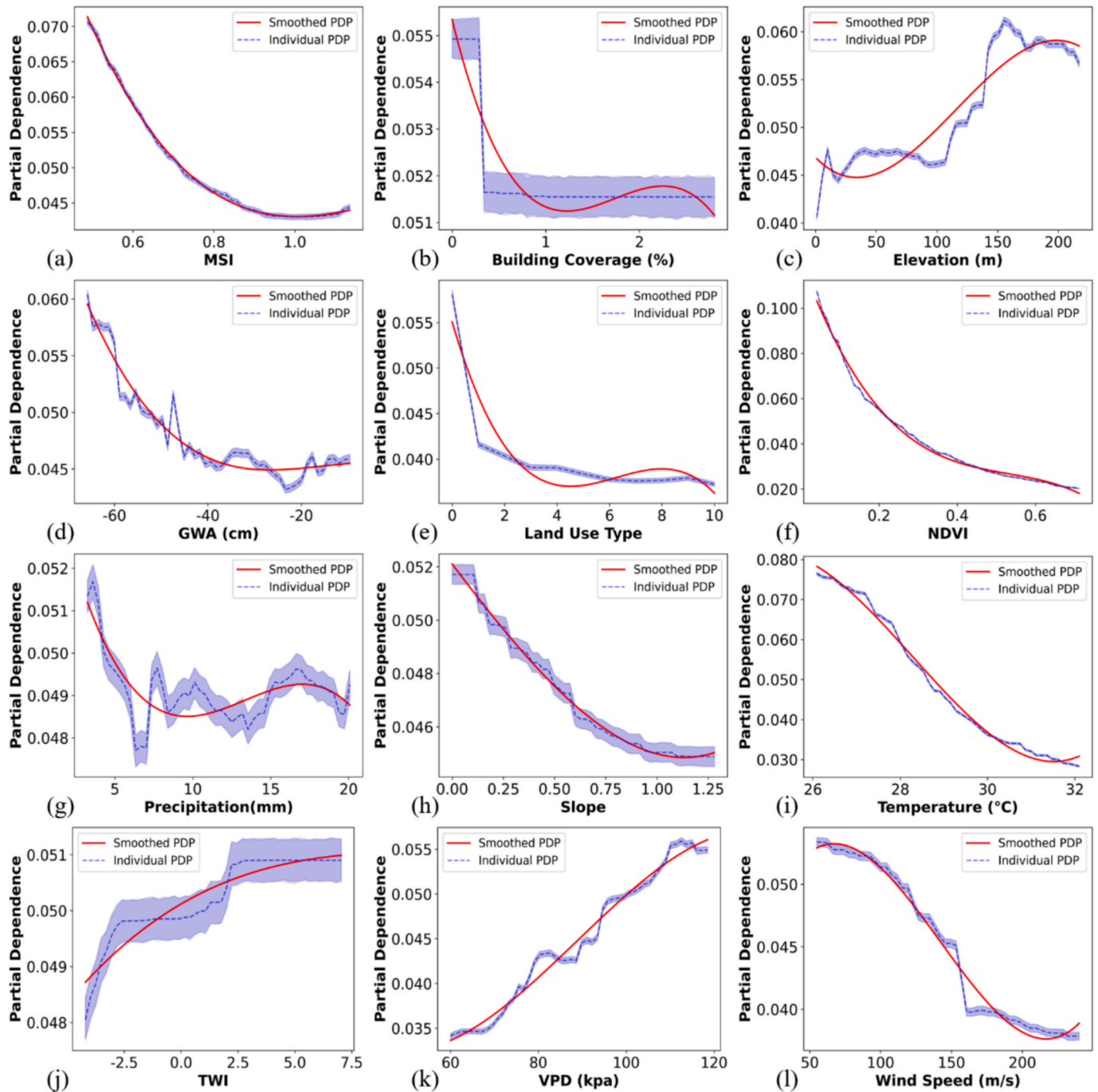


Fig. 12. Partial dependence plots (PDPs) illustrating marginal effects of key hydroclimatic and land-surface variables on the DAIOR predicted by the XGBoost model. The red solid lines represent the smoothed partial dependence functions, while the blue dashed lines denote individual PDPs realizations across samples. The blue shaded regions indicate variability among individual PDPs curves, reflecting model response uncertainty across the data distribution. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

planning, and disaster mitigation.

5.2. Understanding flood drivers from a water cycle perspective

The SHAP and PDPs analyses (Figs. 9 and 12) consistently identify NDVI, temperature, MSI, and land-use type are the dominant predictors of inundation occurrence across the ICP. These results can be interpreted within a water-cycle framework, in which flood occurrence emerges from the partitioning of water among infiltration, evapotranspiration, soil storage, and surface runoff, rather than from precipitation extremes alone.

Specifically, NDVI shows a consistently negative marginal effect on inundation probability, indicating that vegetation-mediated enhancement of soil infiltration and evapotranspiration substantially reduces flood susceptibility in this monsoon-dominated region. Temperature ranks among the strongest predictors, reflecting its dual control on atmospheric moisture demand and soil moisture depletion. During warm seasons, elevated temperatures can reduce soil water storage capacity prior to rainfall events, amplifying runoff responses even under moderate precipitation — an effect supported by recent large-scale attribution studies (Berg and Sheffield, 2018). MSI and land-use type further highlight the importance of surface retention and permeability: areas

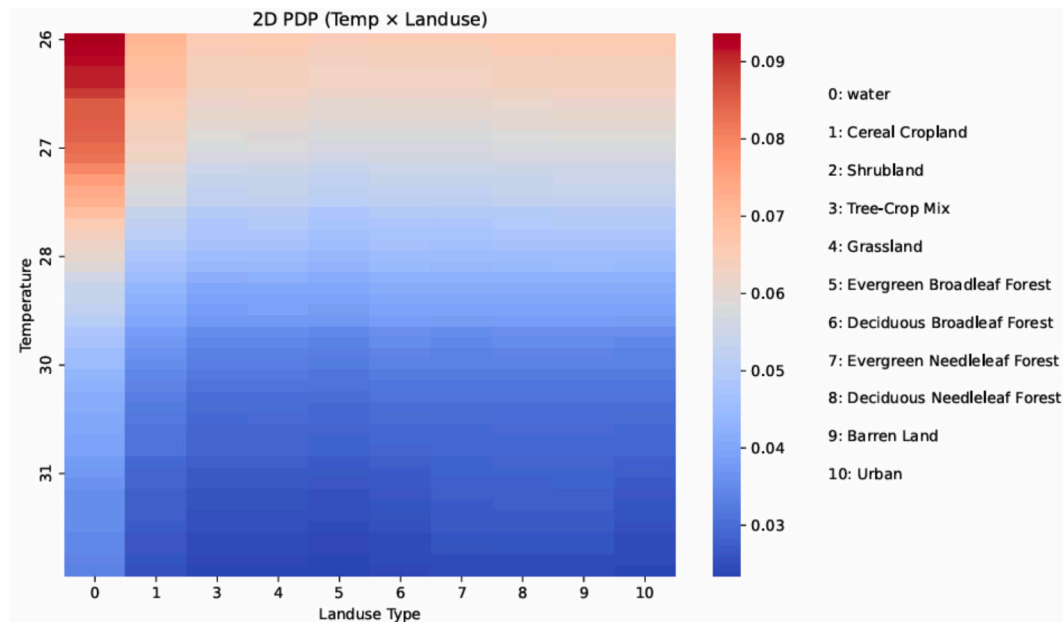


Fig. 13. Two-dimensional partial dependence plot showing the interactive effects of temperature and land use type on flood probability. The interaction between temperature and land use type on the probability of flooding.

characterized by impervious or moisture-stressed surfaces exhibits higher inundation likelihood, whereas agricultural mosaics show strong spatial heterogeneity consistent with observed variations in water retention capacity. Groundwater anomalies further interact with these processes by modulating soil moisture availability.

Collectively, these findings indicate that the flooding in the ICP is not governed solely by rainfall intensity or extremes but arises from the coupled interaction between land-surface and subsurface processes. By embedding model interpretation within a water cycle perspective, this study provides physically consistent evidence that changes in vegetation condition, thermal regime and surface moisture dynamics can substantially modulate inundation occurrence at the regional scale. These results underscore the need to incorporate land surface processes, alongside precipitation, into future flood early warning and land use planning strategies.

5.3. Model comparative analysis with benchmark approaches

To evaluate the effectiveness of different machine learning models in flood susceptibility prediction, we conducted a comprehensive comparison of several algorithms, including XGBoost, RF, SVR, LightGBM, LSTM, and CNN. Each model was trained on the same set of predictors and evaluated using consistent metrics to ensure a fair comparison.

As shown in Table 4, the XGBoost model achieved the highest prediction accuracy ($R^2 = 0.96$, RMSE = 0.014, MAE = 0.0073), which is significantly better than SVR and LightGBM, and is comparable to RF in terms of R^2 . However, XGBoost significantly outperformed RF in terms of computational efficiency, requiring only 29 s and 13441 MB of memory compared to 1939 s and 15238 MB for RF. This highlights XGBoost's strength in both predictive performance and computational efficiency, making it especially suitable for large-scale applications.

In contrast, SVR demonstrated weak predictive capability ($R^2 = 0.33$), indicating its limited capacity to model the complex nonlinear relationships inherent in flood dynamics. LightGBM performed moderately well ($R^2 = 0.92$) but required more computational resources than XGBoost. While deep learning models (LSTM and CNN) exhibited relatively low memory usage (6085 MB and 6213 MB), they did not match XGBoost in predictive accuracy and required longer training times. This difference likely stems from the memory structure of tree-based models,

which store large ensembles of decision nodes, versus the more compact weight matrices used in DL architectures.

Moreover, XGBoost offers practical advantages in interpretability, especially when combined with tools such as SHAP and PDPs. While these methods are applicable to many machine learning models, they are particularly efficient and reliable for tree-based algorithms. This interpretability is especially valuable for flood risk management, where transparency and explanatory insights support evidence-based decision-making.

In summary, among the evaluated models, XGBoost demonstrated a strong balance of predictive accuracy, computational efficiency, and interpretability. These results highlight the practical value of advanced tree-based algorithms for large-scale flood susceptibility modeling in data-scarce and hydrologically complex regions such as the ICP. The comparative findings further emphasize that model selection should consider not only predictive performance but also computational constraints and the capacity to incorporate domain knowledge through interpretable features.

5.4. Application value for regional flood risk management

The 21 years, cross-boundary inundation occurrence data developed in this study provides a unified foundation for regional flood hazard mapping across the ICP. This is particularly valuable for transboundary river systems, such as the Mekong River, where effective flood risk management requires consistent data frameworks to support coordinated planning among countries such as Laos, Cambodia, Thailand, and Vietnam.

In regions where in-situ hydrological monitoring is sparse or unavailable, our remote sensing-driven machine learning framework offers a practical and scalable alternative. By leveraging open-access datasets (e.g., groundwater anomaly, vegetation indices, temperature), the model minimizes reliance on high-resolution, ground-based measurements. This significantly enhances the capacity of low-resource countries to conduct flood exposure assessment and implement early warning systems.

The spatially explicit inundation occurrence maps generated in this study support a range of policy applications, including infrastructure planning, land use planning, and climate adaptation strategies. For

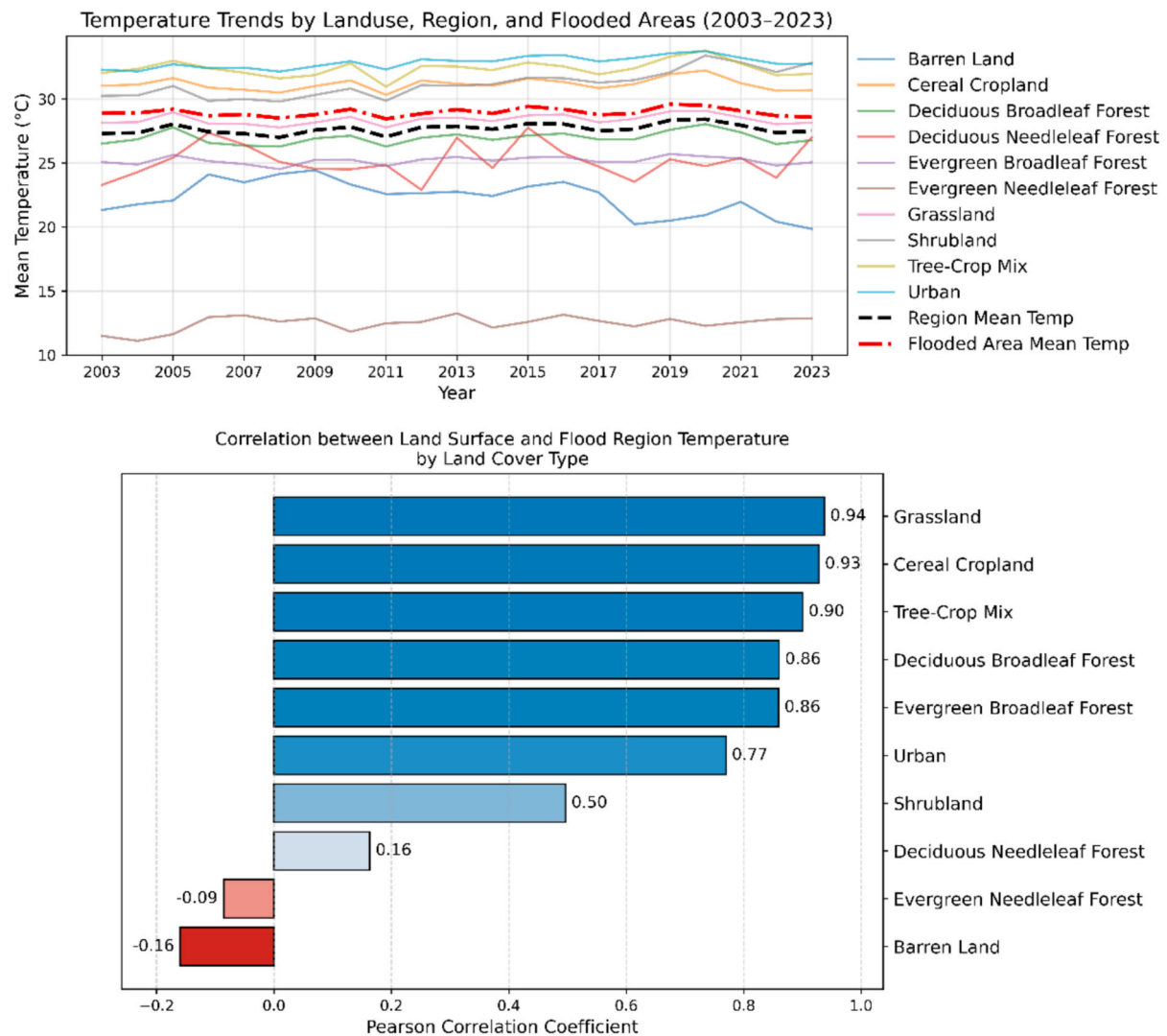


Fig. 14. Temperature trends in temperature across major land-use types (2003–2023) and corresponding time series correlations.

instance, statistical analysis of over two decades inundation in relation to land use type (Fig. 15b) reveal that areas with extensive farmland experience higher flood exposure, while wetlands exhibit strong regulatory effects. Expanding wetland coverage in *peri-urban* zones could therefore enhance landscape-scale flood mitigation and support nature-based solutions. Additionally, the average temperature in the flood-prone areas is higher than the regional baseline (Fig. 14), suggesting that enhanced thermal monitoring and the deployment of temperature sensitive early warning systems may improve preparedness in vulnerable land use zones.

6. Conclusions

This study presents a regional-scale, interpretable machine learning framework for predicting inundation occurrence in transboundary river basins, leveraging two decades of remote sensing data.

By leveraging daily MODIS imagery from 2003 to 2023, the study captures both the spatial distribution and interannual variation of flood events across diverse landscapes. The resulting dataset provides the most internally consistent and continuous record of inundation patterns in ICP to date, filling critical spatial and temporal gaps in existing flood inventories. The use of GEE further enables efficient time series processing and large-scale flood mapping, demonstrating its potential for 21 years dynamic monitoring in data-scarce regions.

The inundation occurrence prediction model, built using XGBoost and 12 selected hydrologically relevant predictors within a water cycle framework, achieved high predictive performance ($R^2 = 0.82$, RMSE = 0.02). Compared with common machine learning and deep learning performance, the proposed XGBoost-based prediction framework has higher prediction accuracy and computational efficiency, which provides a practical solution for large-scale inundation flood susceptibility prediction.

SHAP analysis identified NDVI, land use, and temperature as the most influential predictors. Partial dependence analysis revealed a nonlinear “heat suppression” threshold, where flood risk increases with temperature up to $\sim 30^\circ\text{C}$ but declines thereafter due to enhanced evaporation and reduced soil moisture. Notably, cropland and urban areas exhibited greater sensitivity to temperature extremes, underscoring the role of land surface modifications in amplifying climate-related flood risks. These findings emphasize the importance of integrating land cover feedbacks into climate–flood impact assessments. The proposed methodology is replicable and transferable, offering a robust decision-support tool for flood risk management and sustainable development in vulnerable regions worldwide.

Nonetheless, several limitations remain. First, while MODIS daily images support sufficient temporal resolution, its coarse spatial resolution (250 m) and susceptibility to cloud cover limit its effectiveness in detecting small-scale or canopy-obscured floods. Second, the current

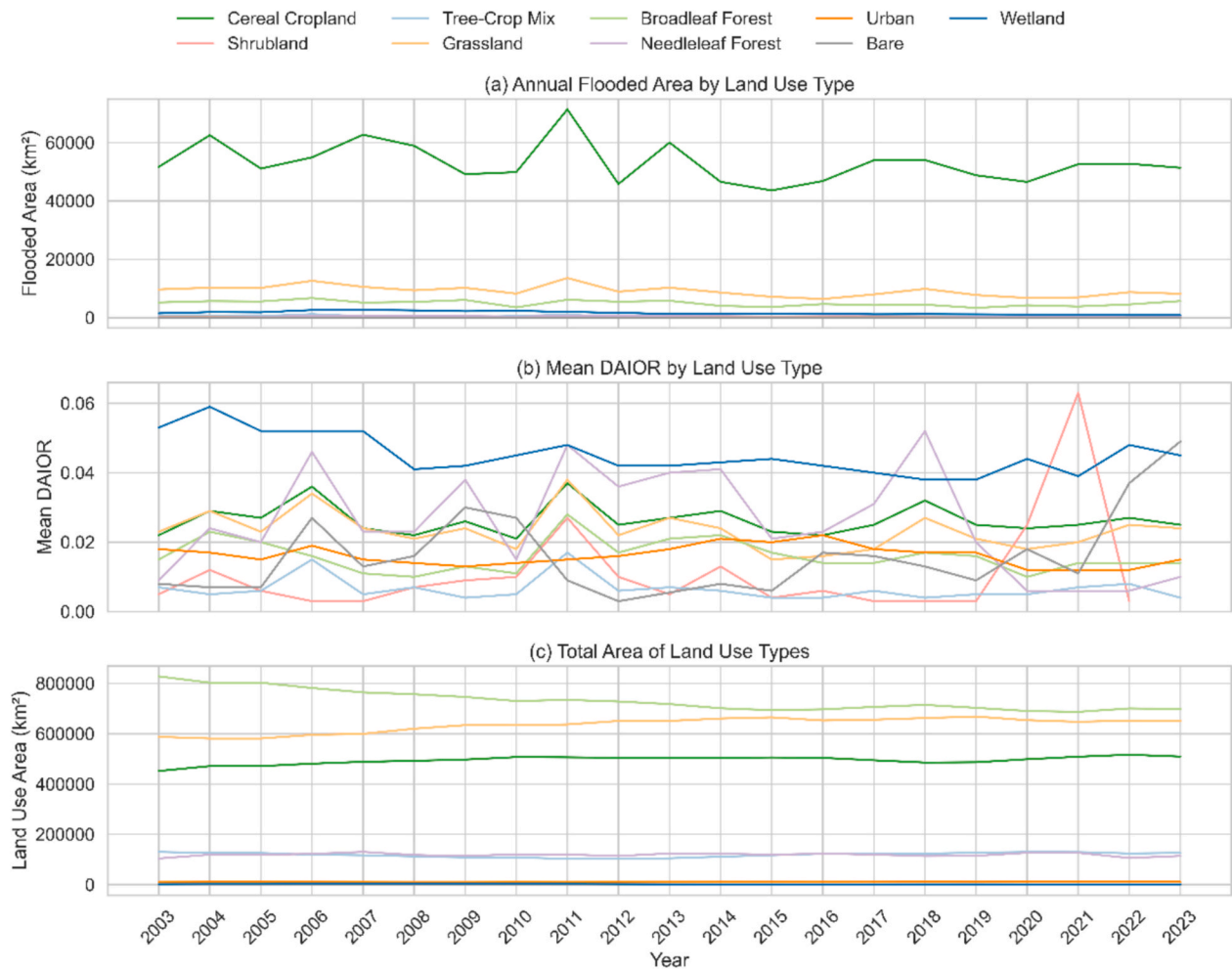


Fig. 15. Temporal trends of flood exposure by land use types in the Ayeyarwady Basin from 2003 to 2023. (a) Annual flooded area, (b) Mean DAIOR, (c) Total land use area.

Table 4

Comparative performance of four machine learning models and two deep learning models in flood susceptibility prediction, evaluated using R^2 , RMSE, MAE, computational time (s), memory usage (MB), and robustness across test sets.

Model	R2	RMSE	MAE	Time	Memory
XGBoost	0.96	0.014	0.0073	29.28	13441.75
RF	0.96	0.018	0.0074	1939.15	15238.67
LightGBM	0.92	0.021	0.0125	10.79	15277.34
SVR	0.33	0.061	0.0530	123105.6	15726.36
LSTM	0.89	0.021	0.0142	520.11	6085.57
CNN	0.88	0.022	0.0144	741.66	6213.20

model focuses primarily on environmental variables; incorporating socio-economic indicators such as infrastructure density and population exposure would enhance its relevance for urban risk assessment. Third, future work should benchmark multiple machine learning and deep learning algorithms to evaluate model robustness across varying climatic and land-use contexts. Fourth, greater attention is needed to address spatial “tail risks”—regions where floods are rare but potentially catastrophic. Improved modeling of these extremes will help capture predictive uncertainty under compounding hydrometeorological and land-surface interactions. Finally, future work will construct a full-chain analysis of flood-exposure-loss. Addressing these challenges will improve the reliability and transferability of data-driven flood models under both historical and future extremes scenarios.

CRediT authorship contribution statement

Gangya Huang: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation. **Alex Hay-Man Ng:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Qianqian Zhou:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Qinggaozi Zhu:** Writing – review & editing, Supervision, Methodology, Investigation. **Zheyuan Du:** Writing – review & editing, Supervision, Investigation. **Linlin Ge:** Writing – review & editing, Supervision, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2026.135176>.

Data availability

The ICP region data is available from GADM maps and data (https://gadm.org/download_country.html). The flood data are available from Global Flood Monitoring System (GFMS) (<https://floodobservatory.colorado.edu/>). Global Hydrography Datasets are derived from MERIT Hydro (Hawke et al., 2020), and are available in Google Earth Engine. Grids of the study area are available from GLANCE Grids (<https://measures-glance.github.io/glancegrids/>).

Global flood dataset was used from Global Flood Database v1 (Tellman et al., 2021), and is available in Google Earth Engine. The MODIS data are available in the USGS EROS Center (<https://lpdaac.usgs.gov/products/mod09gqv006/>), (<https://lpdaac.usgs.gov/products/mod09gav006/>), and are mirrored in the Google Earth Engine data catalogue. The Sentinel-1 SAR GRD products used for the accuracy assessment provided by the European Union Copernicus (<https://sentiwiki.copernicus.eu/web/slapplications>) and are available in the Google Earth Engine data catalogue.

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